Customer Monitor on SAP BTP – Solution Design (Design Review 10.11.2026)

Field	Value
Document	Customer Monitor on SAP Business Technology Platform – solution design and answers to the questions for the design review
Status	Draft for the design review
Version	1.2 – adds the summary, the delivery status, the validation caveats and the requirement index; 1.1 took HYDAC's BTP usage export of October 2026 (`Subaccount_Details.xlsx`) into account
Date	2026-11-09
Author	Igor Vorobev, purplecat AG (SAP BTP solution architecture)
Related	Kundenmonitor (ZEC_CUSTOMER_MONITOR) – Technische und fachliche Dokumentation (legacy report analysis) · Australien-Konzeption: Integration Architecture (shop integration, ADR-022/023) · Ergebnisprotokoll Schnittstellen-Workshop BTP (06.–07.10.2026)

Summary and recommended approach. The Customer Monitor replaces the regional SAP GUI review of shop registrations with one browser-based work queue on SAP BTP. Registrations from the Magento shops are received and stored on BTP, reviewed by the back office and processed against the SAP system of the request's region. The monitor owns the requests and their workflow; the SAP systems stay the system of record for business partner and customer master data and are reached only through the standard API `API_BUSINESS_PARTNER`, so there is nothing to develop in S/4HANA on-premise or in the Australian Public Cloud.

We propose one SAP Fiori Elements application on a SAP CAP service with SAP HANA Cloud, deployed as one archive. Development starts in a new space of HYDAC's existing

`DEV` subaccount, using the central launchpads, the shared HANA Cloud instance and the cTMS tenant that HYDAC already operates (chapters 5.3 and 7). For production we recommend one central deployment for all regions, on the condition that HYDAC data protection approves EU-hosted Australian contact data; otherwise Australia is deployed regionally from the same archive (section 7.3).

How to read this document. The document follows the arc42 structure, like the Integration Architecture page. The four points of Peter Wurst's email and the topics of the meeting invite are answered in these chapters:

Topic	Chapter
Functional design: business purpose, users, use cases, the data the monitor displays and acts on	1, 3.4, 5.2, 6.1
Requested elements mapped to requirements: for each entitlement, role, space, Git and transport item, which requirement it supports and why it is needed	5.3, 7.2, 7.4, 7.5, 10.1
Ongoing administration: the types of administration and maintenance responsibilities of the application and its BTP services, so HYDAC can decide who owns each one	8.2, 8.4, 8.7, 8.8
Integration: connection to Q4H / GWQ and later production; HYDAC Australia's S/4HANA Public Cloud; Magento; Cloud Connector resources; OData services / RFC modules / APIs; authentication; relationship to the Integration Suite iFlows	3, 5.4, 5.5, 6, 7.1, 7.4, 8.1, 8.9
Invite topics: required BTP services · connection setup · multi-backend architecture · roles and authorizations · user management	5.3 · 7.4, 8.1 · 7.3, 8.3 · 1.4, 8.1 · 8.2

Status labels used throughout:

Label	Meaning
✅ Built	Built and tested against mock data (137 service tests, 15 UI journeys).
🟡 Designed	Designed from SAP's documentation, not built yet; integration or authentication mechanisms marked 🟡 have **not yet been validated** in HYDAC's systems (section 10.2 names what the first connectivity tests have to confirm).
💡 Proposal	Suggested value or approach; needs agreement with HYDAC.
❓ Open	Open point for HYDAC; collected in chapter 11.
Assumption	Our current understanding, please correct.

Identifiers: `UC-nn` use case (section 3.4), `REQ-x` requirement (section 10.1), `I-nn` interface (section 3.3), `D-nn` decision (section 4.2), `ADR-nn` architecture decision (chapter 9).

> **Sources** named in the text: Peter Wurst's email of 30 October 2026 and the meeting invite of 4 November 2026; HYDAC's BTP usage export of October 2026 (`Subaccount_Details.xlsx`); the analysis of the legacy report `ZEC_CUSTOMER_MONITOR` and its custom objects (Confluence page "Kundenmonitor (ZEC_CUSTOMER_MONITOR) – Technische und fachliche Dokumentation"); the SAP documentation of `API_BUSINESS_PARTNER` , SAP Cloud Connector principal propagation and the communication scenario `SAP_COM_0008` .

1. Introduction and Goals

1.1 Task

HYDAC's Magento B2B shops send three kinds of input to SAP: **company registrations** (new companies, or companies that already exist as a customer – Debitor), **new shop users** (contact persons – Ansprechpartner – for an existing customer) and **new delivery addresses** (ship-to parties – Warenempfänger – for an existing customer). Today a custom SAP GUI report (`ZEC_CUSTOMER_MONITOR`) with custom tables, custom OData services and custom step classes in S/4HANA on-premise lets the back office review these requests and create the business partner, customer, contact person and ship-to party. The report serves one SAP system. HYDAC now needs the same process for Europe, the USA and Australia: Australia runs

SAP S/4HANA Cloud Public Edition, and the USA will get an own S/4HANA system on RISE with SAP (S/4HANA Cloud Private Edition) in the future.

The Customer Monitor is the human review step between the shop and SAP master data. Neither today nor in the target does the intake create master data directly. Each input is stored as a **request** with the original payload and a working copy, and an approval check runs. The back office reviews and completes the request in the monitor and then triggers **steps** that create or assign the business partner, customer, contact person or ship-to in the SAP system of the region. The monitor reports the result back to Magento.



Diagram source (Mermaid)

01-process-overview.mmd

```
flowchart LR
  A["Shop user registers<br/>company · contact · address"] --> B["Magento sends<br/>regist
  B --> C["Request stored<br/>process type determined<br/>approval check"]
  C -->|release required| D["Releaser approves<br/>or rejects"]
  C -->|released automatically| E
  D --> E["Back office reviews,<br/>corrects data"]
  E --> F{"Step"}
  F -->|CREATE / CREATE_WE / AP_MODIFY| G["Business partner · customer ·<br/>contact · shi
  F -->|CHNG_OLD / CNG_DEBI / CNG_NEW| E
  F -->|REJECT / MANUAL_OK| H["Request closed"]
  G --> I["Magento notified<br/>(IDs or rejection)"]
  H --> I
```

Input from Magento	Process type	Result in SAP
Company registration, no SAP customer yet	`CR_DEB`	Business partner with customer role, company code and sales area data; admin contact person
Company registration, customer exists (duplicate handling)	`CH_DEB`	Contact person created or changed on the existing customer
New shop user for an existing customer	`CR_AP` / `CH_AP`	Contact person created (`CR_AP`) or changed automatically (`CH_AP`)

New delivery address for an existing customer	`CR_WE`	Ship-to party as customer, partner function on the sold-to

1.2 Goals

#	Goal	Explanation
G1	One work queue for all regions	Browser access via SAP Build Work Zone, one app for EU, USA and AU, deep links (for example in emails); tabs with counters and a "next step" recommendation per request instead of one SAP GUI list per system
G2	No custom code in the SAP systems	The SAP systems serve the standard business-partner API only; the `ZEC_*` tables, customizing, step classes, Gateway services and BAdI are retired. Australia's Public Cloud needs no ABAP Cloud development
G3	Correct processing under concurrency and failure	Edit lock, version check and serialized steps per request; numbers created in SAP are persisted at once, so a repeated step never creates a second business partner
G4	One authorization model	One sales-org rule in the service for the list, the detail and all sub-data; S/4 checks the real user when master data is created
G5	Robustness by design	The defects found in the legacy report (lost ship-to

		numbers, notifications before the commit, swallowed send errors, release limit without currency) are fixed in the service, with nothing depending on changes in S/4
G6	Business rules as data	Allowed steps, approval rules and texts per region are maintained by key users in the application, without deployment or transport

Business value of the new monitor compared with the SAP GUI report:

Area	Legacy (SAP GUI)	New (BTP / Fiori)
Access	SAP GUI, transaction, one system	Browser via SAP Build Work Zone, one app for all regions, deep links (for example in emails)
Work queue	One list with filters	Tabs with counters: My work, Errors, Clarification, Release, All; sorted by priority and waiting time; "next step" recommendation per request
Concurrency	Steps run without a lock, so two users can create the same customer twice	Edit lock (draft), version check (ETag) on every step, steps serialized per request. The request, log and outbox are written in one database transaction; the call to SAP is a separate operation whose returned numbers are persisted at once, so a repeat continues where the step stopped

Authorization	Two different sales-org checks	One sales-org rule in the service for the list, the detail and all sub-data; S/4 checks the real user when master data is created
Robustness	Several known defects found in the analysis of the legacy report (lost ship-to numbers after partial failures, notifications sent before the commit, swallowed send errors, release limit compared without currency)	Fixed by design in the service; nothing depends on changes in S/4
Custom code in SAP	Tables, customizing, step classes, Gateway services, BAdI (`ZEC_*`, `ZCL_EC_*`)	**None.** The SAP systems serve the standard business-partner API only

1.3 Quality Goals

Priority	Quality goal	Scenario / measure
1	Correctness of master data	A step that fails half-way (for example the contact person after the business partner) keeps the created numbers on the request; the repeat continues with the missing part. A stale version returns 412 instead of overwriting. Verified by service tests
2	Security and traceability	A user without a sales-org assignment sees nothing (fail closed); master data is created on the real user via principal propagation, so S/4 authorizations and change

		documents apply. Every field change and every step is in the request log
3	Isolation and availability	If one SAP system is unreachable, only that region's master-data steps fail with `BACKEND_UNAVAILABLE`; everything else keeps working. Notifications to an unreachable shop wait in the outbox and are retried
4	Maintainability	A new region is one registry entry plus two destinations; customizing is data; the same archive moves through DEV, QAS and PRD with stage-specific values only in extension descriptors
5	Testability	137 service tests and 15 UI journeys run in the CI/CD pipeline before every deployment; the step logic is tested against mock backends

1.4 Stakeholders

Organisation	Role in this solution
HYDAC Group IT (BTP, SAP Basis, identity, integration)	Operates the platform; owns identity, security, transport approvals and production; decides the open points of chapter 11
HYDAC back office EU, USA, AU	Users of the monitor: review, correct and process requests per sales organization
HYDAC regional key users	Administer customizing and sales-org assignments for their region (role

	`Kundenmonitor_Admin` 💡); first-level support
HYDAC Australia IT	Runs the same process on the S/4HANA Cloud Public Edition tenant without ABAP development (section 5.4)
HYDAC Security	Decides data residency (section 7.3) and retention (section 8.6)
TechDivision (Magento)	Switches the shop endpoints at cutover and keeps the payload contract (section 5.5)
purplecat AG	Builds, deploys and supports the application

Users and their roles (BTP role collections):

Role (BTP role collection)	Who	May	Legacy equivalent
`Kundenmonitor_Viewer`	Sales / customer service staff who only need to look up requests	Read requests of their own sales organizations	Display
`Kundenmonitor_Editor`	Back office (Innendienst) per sales organization, in EU, USA, AU	+ edit request data, run steps	Change
`Kundenmonitor_Releaser`	Users who approve requests that need a release (Freigabe)	+ see unreleased requests and release them	Authorization object `ZEC_MONITO`, activity 43
`Kundenmonitor_Admin`	Regional key users	Maintain customizing (steps, approval rules, texts) and sales-org assignments for their region	Customizing transactions

The roles define **what** a user may do. **Which** requests they see is defined by the sales organizations assigned to them per backend (region), enforced in the service on every read (section 8.1). A user without an assignment sees nothing (fail closed). The same role collections serve all regions; there are no region-specific roles.

User	Roles	Backend: sales orgs	Sees
Back office Germany	Editor	EU: 0001, 2000	Open EU requests of 0001/2000, can run steps
Back office USA	Editor, Releaser	US: xxxx	US requests including unreleased ones, can release
Back office Australia	Editor	AU: xxxx	AU requests only
Global support	Viewer	all backends, all sales orgs	Everything, read only

❓ Number of users per role, region and sales organization, and the request volume per day per region. We need these for sizing and for the user administration effort.

2. Constraints

#	Constraint	Consequence
C1	HYDAC Australia runs S/4HANA Cloud Public Edition	No ABAP, no custom OData services, no developer extensibility. Customers can only be created from outside through `API_BUSINESS_PARTNER` (the released ABAP Cloud business-partner object cannot create customer roles, SAP note 3655149). The same API is therefore used for all regions

C2	The SAP systems stay the system of record for business partner and customer master data	The monitor owns the requests and their workflow only; master data is created through the standard API on the real user and never read remotely for the work queue
C3	The Magento payload contract stays as it is today	TechDivision changes only endpoint and credentials at cutover; the intake accepts the structure Magento sends to `ZEC_CUSTOMER_CREATE_SRV` today
C4	HYDAC's global account has subaccounts only in Europe (eu10 on AWS, eu20 on Azure); HYDAC USA's accounts-payable automation already keeps its data in eu10	A central deployment for EU and USA follows existing practice; a regional AU deployment would be HYDAC's first subaccount outside Europe
C5	Australian Privacy Principle 8 restricts cross-border disclosure of personal information	The data residency decision for AU contact data is needed before AU goes live, not before DEV (section 7.3)
C6	HYDAC operates central launchpads (`CFLPDEV` , `CFLPPRD`) and a cTMS tenant (`ctms-hydac` in subaccount `CTMS`)	The monitor needs no Work Zone and no cTMS subscription of its own: content provider in the launchpad subaccounts, transport nodes in `CTMS`
C7	The `DEV` subaccount runs the Cloud Foundry Free plan	The monitor needs the Standard (MEMORY) plan quota of 2–4 GB in `DEV` ❓
C8	Integration Suite's built-in Git integration supports	HYDAC-owned GitHub organization with the repositories `cap-monitor`

	github.com only and covers iFlows only	and `cpi-iflows`; message mappings and script collections stay in Integration Suite packages
C9	Requests contain personal data of contact persons; from phase 2 HANA Cloud on BTP is the only copy	CAP `@PersonalData`, SAP Audit Log service 💡, scheduled retention job, data residency (section 8.6)
C10	The known defects of the legacy report must not be carried over	Lock, version check, serialized steps, outbox after commit and currency-aware release limit are part of the service design (section 8.5)
C11	The USA will move to an own S/4HANA system on RISE with SAP (S/4HANA Cloud Private Edition)	Backend `US` is a separate entry in the backend registry from the start; the RISE system is reached like the EU system (Cloud Connector in the RISE landscape, `API_BUSINESS_PARTNER`, principal propagation), so the migration is a destination change, not a code change. ❓ Timeline, and which system serves US registrations until then (section 7.1)

3. Context and Scope

3.1 Business and Technical Context

Principle: one UI, one service, one request store, one step logic; per region one SAP destination and one Magento destination. The requests of all regions are stored in the monitor; the master data goes to the region's SAP system.

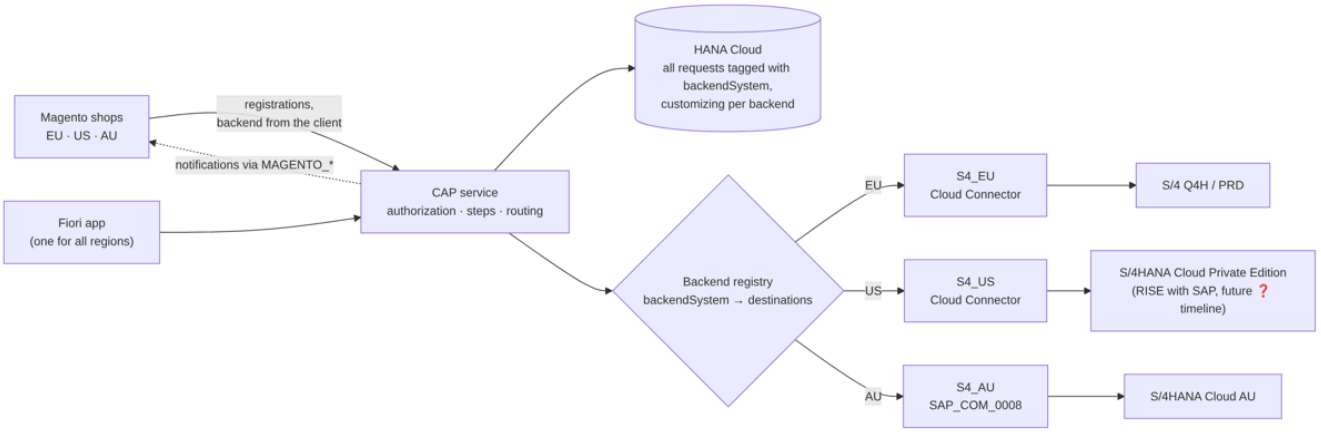


Diagram source (Mermaid)

08-multi-backend.mmd

```
flowchart LR
  MAG["Magento shops<br/>EU · US · AU"] -->|"registrations,<br/>backend from the client"| C
  UI["Fiori app<br/>(one for all regions)"] --> CAP
  CAP --> H[("HANA Cloud<br/>all requests tagged with backendSystem,<br/>customizing per ba
  CAP --> REG{"Backend registry<br/>backendSystem → destinations"}
  REG -->|EU| D1["S4_EU<br/>Cloud Connector"] --> B1["S/4 Q4H / PRD"]
  REG -->|US| D2["S4_US<br/>Cloud Connector"] --> B2["S/4HANA Cloud Private Edition<br/>(RI
  REG -->|AU| D3["S4_AU<br/>SAP_COM_0008"] --> B3["S/4HANA Cloud AU"]
  CAP -.->|"notifications via MAGENTO_*"| MAG
```

Partner	Role	Communication
Magento shops EU · US · AU (TechDivision)	Source of registrations, new shop users and delivery addresses; receive ID notifications and rejections	Intake: REST/OData V4 over HTTPS, OAuth 2.0 client credentials per shop (I-01). Notifications: HTTPS via destination per shop, credentials as today ❓ (I-03)
S/4HANA on-premise EU (Q4H via GWQ, later the production pair ❓)	System of record for business partner, customer, contact person, ship-to	`API_BUSINESS_PARTNER` (OData V2) via destination, Cloud Connector and principal propagation (I-02)
S/4HANA USA – future own system on RISE with SAP (S/4HANA Cloud Private Edition)	System of record for the US master data once migrated; ❓ which system serves US registrations until then	`API_BUSINESS_PARTNER` via destination `S4_US` and the Cloud Connector of the RISE landscape, principal propagation – the same mechanism as EU (I-02)

S/4HANA Cloud Public Edition AU (HYDAC Australia)	System of record for the AU master data	`API_BUSINESS_PARTNER` via the communication arrangement `SAP_COM_0008`, OAuth 2.0 SAML bearer assertion (I-02)
Corporate identity provider (❓ Entra ID) and SAP Cloud Identity Services	Identity and role groups of the business users	SAML / OIDC trust to the BTP subaccount; groups mapped to role collections
SAP Build Work Zone, standard edition (`CFLPDEV`, `CFLPPRD`)	Entry point for business users	Content provider from the launchpad subaccount to the application subaccount; managed approuter forwards the user token
GitHub (HYDAC org), SAP CI/CD, Cloud Transport Management	Delivery of the application	Webhook → build → `cf deploy` to DEV, archive upload to cTMS, imports approved by HYDAC
SAP Integration Suite (`FSM-DEV-CPI`, `FSM-PRD-CPI`)	Not in the runtime path; optional API Management proxy in front of the intake, legacy Magento iFlows	Section 8.9

3.2 Interaction Patterns

Pattern	Direction	Mechanism	Used for
Synchronous user call	Fiori app → CAP service	OData V4 with the user's token, forwarded by the managed approuter; drafts, ETag, actions	Work queue, request review, editing, all steps (UC-01 to UC-13)
Synchronous intake	Magento → CAP IntakeService	REST over HTTPS, OAuth 2.0 client credentials (one XSUAA service key	I-01: company registration, new shop user, new delivery address

		per shop/region); idempotent on the Magento IDs	
Synchronous master-data call	CAP service → SAP system of the request	`API_BUSINESS_PARTNER` (OData V2) through the destination and connectivity services with user propagation; result persisted at once	I-02: steps CREATE, CREATE_WE, AP_MODIFY; duplicate check; value help
Asynchronous notification	CAP outbox worker → Magento	Persistent outbox written in the same transaction as the step, sent after the commit, retried with backoff, at least once	I-03: ID notification, rejection
Local read	Fiori app → HANA Cloud	All lists, counters, details and value helps are served from HANA Cloud; no remote reads, no synchronization	Work queue across all regions, filtered by the user's assignments

3.3 Interface Catalogue

Three interfaces replace the six custom objects of today (`ZEC_CUSTOMER_CREATE_SRV` , `ZEC_CUSTOMER_READ_SRV` , `RejectCusomterSet` , `ZEC_CUSTOMER*` tables, `ZCL_EC_CUSTOMER` step classes, BAdI notification). **No RFC function modules, no IDoc, no custom OData service, no ABAP.** The service calls the standard API through SAP's destination and connectivity services (SAP Cloud SDK underneath).

ID	Interface	Direction	Protocol / authentication	Content	Replaces	Status

I-01	**Intake:** company registration, new shop user, new delivery address	Magento → BTP	REST/OData V4 over HTTPS, OAuth 2.0 client credentials (one XSUAA service key per shop/region); optional API Management proxy (section 8.9)	Today's payload structure; creates the request, runs the approval check, writes the intake log entries; idempotent on the Magento IDs	`ZEC_CUSTOMER_CREATE_SRV`	🟡 Designed
I-02	**Master data:** steps CREATE, CREATE_WE, AP_MODIFY; duplicate check (CHNG_OLD, CNG_DEBI); value help	BTP → SAP	**Standard** `API_BUSINESS_PARTNER` (OData V2) via destination: Cloud Connector + principal propagation (on-premise), communication scenario `SAP_COM_0008` + OAuth 2.0	Business partner with address, roles, tax numbers; customer with company code and sales area; contact person; ship-to and partner function; search by number/name	CMD-API / BAPI calls in `ZCL_EC_CUSTOMER*`	✅ step logic, 🟡 API call

			SAML bearer (Public Cloud)			
I-03	**Notifications:** ID notification after a successful step, rejection	BTP → Magento	HTTPS via destination per shop, credentials as today (❓ TechDivision); persistent outbox, sent only after the commit, retried, result logged	Same targets and payloads as today (`RejectCustomerSet` kept; address ID for `CR_WEB` rejections to agree)	BAdI notification, `ZEC_CUSTOMER_READ_SRV` read-back	🟡 Designed
–	Legacy Magento ↔ S/4 custom services	–	–	Retired per shop at cutover after a parallel run (section 5.5)	–	–

3.4 Use Cases

ID	Use case	Role	Status
UC-01	**Work the queue:** views My work / Errors / Clarification / Release / All with counters, sorted by	Viewer+	✅

	priority and waiting time, restricted to the user's sales organizations		
UC-02	**Review a request:** main data, contact person or ship-to, sales data, assignment (SAP and Magento numbers, release reason), processing log	Viewer+	✅
UC-03	**Correct or complete data** before processing (draft with lock; every field change logged)	Editor	✅
UC-04	**Release** a request that needs approval (rule per process type and sales org: manual, credit card, order-volume limit)	Releaser	✅
UC-05	**Create customer and admin contact** (step CREATE)	Editor	✅ logic, 🟡 API call
UC-06	**Create or change contact person** (AP_MODIFY)	Editor	✅ logic, 🟡 API call
UC-07	**Create ship-to party** and assign it to the sold-to (CREATE_WE)	Editor	✅ logic, 🟡 API call
UC-08	**Handle duplicates:** assign an existing	Editor	✅

	customer (CHNG_OLD), change the assignment (CNG_DEBI, with a confirmation question if the customer belongs to another Magento company), convert to a new customer (CNG_NEW)		
UC-09	**Put in clarification** (Klärung) with a mandatory comment	Editor	✅
UC-10	**Reject** with a mandatory reason; Magento notified	Editor	✅ logic, 🟡 notification
UC-11	**Complete manually;** reset a closed request for reprocessing	Editor	✅
UC-12	**Fix errors:** a failed step sets an error status with the SAP messages in the log; the user corrects the data and repeats the step. Numbers already created in SAP are kept, so a repeat never creates a second business partner	Editor	✅ logic, 🟡 API call

UC-13	**Maintain customizing** and user assignments per region in the application	Admin	💡
UC-14	**Intake of registrations** from Magento (three operations), idempotent	Magento (technical client)	🟡
UC-15	**Migrate open requests** and Magento mappings from `ZEC_CUSTOMER` at cutover	purplecat / HYDAC	🟡

Which steps are allowed is decided **by the service** (customizing per backend plus role). The UI only shows what the service allows; it holds no business rules.

Delivery status at the time of this review. The work queue, request review and editing, release, duplicate handling, clarification, rejection, manual completion, reset and the complete step logic are built and tested against mock data (137 service tests, 15 UI journeys). Not yet built: the calls to `API_BUSINESS_PARTNER`, the Magento intake and notifications, and the migration of open requests (phase 2, after the decisions of this review). The admin screen is a proposal. A live demo of the current version is possible in the session.

3.5 Scope

In scope	Out of scope	Later / optional
Review and processing of the three request kinds for all process types, statuses and steps of the legacy report	Changes to the Magento shops beyond endpoint and credentials	API Management proxy in front of the intake (section 8.9)
Intake from Magento, approval check, notifications to Magento	Replacing the S/4 BAdI notification on customer maintenance with partner role	Business-partner events S/4 → BTP

	ZO (business-partner events, separate work package)	
Master-data creation through `API_BUSINESS_PARTNER` in S/4 on-premise and S/4HANA Cloud Public Edition	Master-data maintenance beyond what the steps create (done in S/4 apps)	Navigation from the monitor to the S/4 app "Manage Business Partner Master Data"
Multi-region operation: EU, USA, AU with one deployment or one per region	Workflow / approval outside the monitor (SAP Build Process Automation)	Work Zone integration of further apps
Customizing per region in the application, administered by key users		Retention job for completed requests, audit log of personal-data access 💡
Cutover: migration of open requests and Magento mappings, retirement of the custom S/4 objects		

4. Solution Strategy

4.1 Core Principles

The application is a SAP Fiori Elements user interface on a SAP CAP (Node.js) service with SAP HANA Cloud, deployed as one archive into a Cloud Foundry space. The design rests on these principles:

- **BTP is the system of record for the requests;** the SAP systems are the system of record for master data and are reached only through the standard API `API_BUSINESS_PARTNER`. No custom code in any SAP system, on-premise or Public Cloud.
- **One UI, one service, one request store, one step logic for all regions;** per region one SAP destination and one Magento destination, resolved through a backend registry (configuration, not code).
- **The service decides, the UI shows.** Allowed steps, release checks and process-type changes are customizing data per backend, evaluated in the service; the Fiori Elements UI holds no business rules.
- **Reads are local, writes are transactional.** Lists, counters and details come from HANA Cloud; every step changes the request, log and outbox in one transaction, and numbers

returned by SAP are persisted at once.

- **The real user creates master data.** Principal propagation to the SAP system, so S/4 authorizations and change documents apply to the person who ran the step.
- **Identity and roles come from HYDAC's identity provider;** the data scope (sales organizations per backend) is maintained by regional key users in the application.
- **One archive through all stages.** DEV → QAS → PRD via cTMS; stage-specific values only in extension descriptors; customizing travels as data, not with the archive.

4.2 Decision Log

#	Decision	Status	Source
D-01	Develop in a new Cloud Foundry space `cap-monitor` of HYDAC's existing `DEV` subaccount (eu10); the question about `FSM-DEV-CPI` is withdrawn, that subaccount is Integration Suite only	💡 Proposal, HYDAC decides	Section 7.2, BTP usage export
D-02	Map the existing HANA Cloud instance in `GeneralServices` to the monitor's DEV and QAS spaces instead of a new `hana-cloud` entitlement	💡 Proposal · ❓ capacity and policy	Section 5.3
D-03	One central production deployment for all regions, conditional on HYDAC data protection approving	💡 Proposal · ❓ Open (needed before AU go-live)	Section 7.3

	EU-hosted AU contact data; otherwise regional for AU		
D-04	Integrate the monitor into HYDAC's central launchpads `CFLPDEV` / `CFLPPRD` as a content provider; no Work Zone subscription in the application subaccount	💡 Proposal · ❓ confirm HYDAC standard and the QAS site	Section 5.3
D-05	Reuse the cTMS tenant `ctms-hydac` in subaccount `CTMS` with one transport node per target space	💡 Proposal · ❓ confirm owner	Section 7.5
D-06	Principal propagation for the master-data steps; technical user only for the first connectivity tests on DEV, principal propagation before user tests on QAS	💡 Proposal · ❓ HYDAC decides	Section 8.1
D-07	Roles via IAS groups → role collections; data scope in the application table `UserAssignments`, maintained by regional key users. Writing the scope as an IAS user attribute via SCIM rejected as the default	💡 Proposal	Section 8.2

D-08	The Magento mapping (company / customer / address ↔ SAP numbers) lives in HANA Cloud, written by the steps and migrated at cutover; no Z field in S/4	💡 Proposal · ❓ key-user field if S/4 users need it	Section 6.3
D-09	Notifications to Magento through a persistent outbox, sent after the commit, retried, delivered at least once	🟡 Designed · ❓ Magento tolerates repeats	Section 6.4
D-10	The monitor's runtime calls do not go through Integration Suite; an API Management proxy in front of the intake is optional	💡 Proposal · ❓ HYDAC API policy	Section 8.9
D-11	Completed requests are deleted by a scheduled retention job after 24 months	💡 Proposal · ❓ period to be defined by HYDAC	Section 8.6
D-12	Build with SAP CI/CD (alternative: GitHub Actions + `cf` CLI)	❓ Open: check whether SAP Build Code already covers the CI/CD service	Section 7.5
D-13	Customizing (steps, approval rules, texts) per backend as data in HANA Cloud,	💡 Proposal	Section 8.4

	maintained in an admin screen (UC-13)		
D-14	Sales-org assignments carry the backend, because the same code can exist in two SAP systems with a different meaning	💡 Proposal · ❓ plain codes suffice if codes are unique	Section 8.1
D-15	The USA is modelled as its own backend `US` from the start, pointing at the future RISE system (S/4HANA Cloud Private Edition) with the same adapter as EU; until the migration the destination points at the interim system	💡 Proposal · ❓ timeline and interim system	Section 7.1

5. Building Block View

5.1 Overview

The picture shows where each BTP service is used:

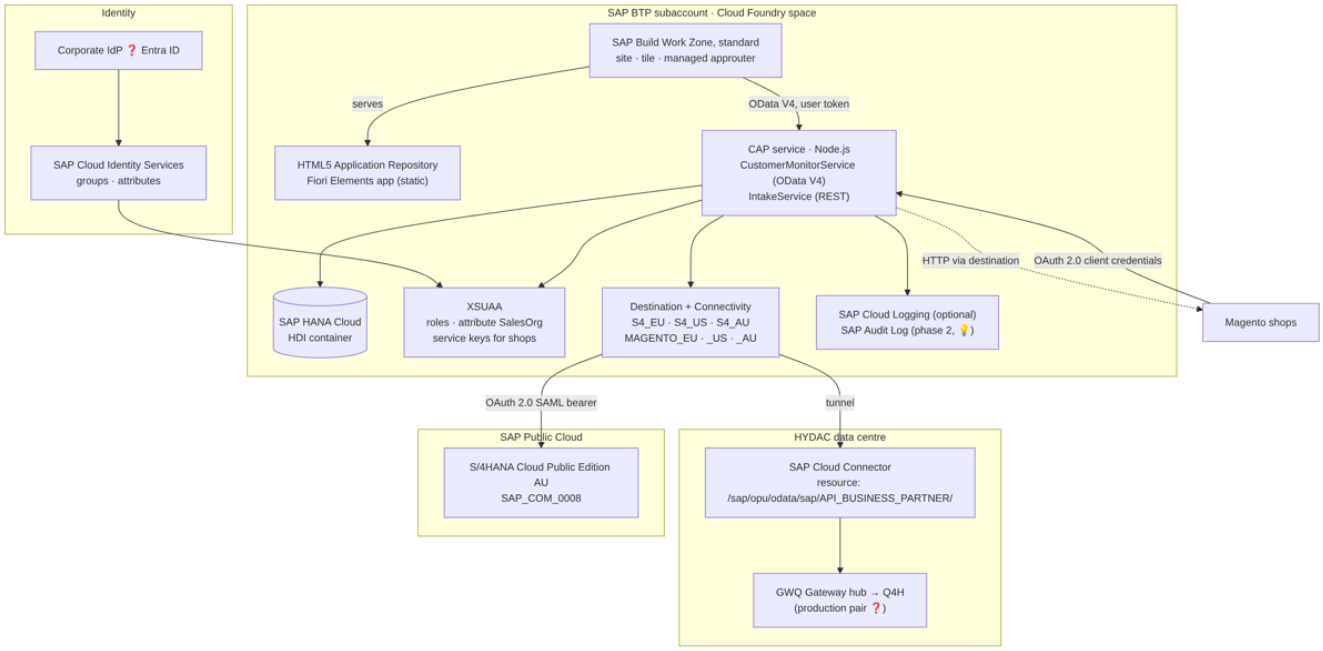


Diagram source (Mermaid)

04-btp-services.mmd

```
flowchart TB
  subgraph IDENT["Identity"]
    IDP["Corporate IdP ❓ Entra ID"]
    IAS["SAP Cloud Identity Services<br/>groups · attributes"]
  end
  subgraph SUB["SAP BTP subaccount · Cloud Foundry space"]
    WZ["SAP Build Work Zone, standard<br/>site · tile · managed approuter"]
    HTML5["HTML5 Application Repository<br/>Fiori Elements app (static)"]
    SRV["CAP service · Node.js<br/>CustomerMonitorService (OData V4)<br/>IntakeService (RE
    HDI[("SAP HANA Cloud<br/>HDI container")]
    XS["XSUAA<br/>roles · attribute SalesOrg<br/>service keys for shops"]
    DST["Destination + Connectivity<br/>S4_EU · S4_US · S4_AU<br/>MAGENTO_EU · _US · _AU"]
    OBS["SAP Cloud Logging (optional)<br/>SAP Audit Log (phase 2, 💡)"]
  end
  subgraph ONPREM["HYDAC data centre"]
    CC["SAP Cloud Connector<br/>resource: /sap/opu/odata/sap/API_BUSINESS_PARTNER/"]
    GWQ["GWQ Gateway hub → Q4H<br/>(production pair ❓)"]
  end
  subgraph PUB["SAP Public Cloud"]
    PC["S/4HANA Cloud Public Edition AU<br/>SAP_COM_0008"]
  end
  MAG["Magento shops"]
  IDP --> IAS --> XS
  WZ -->|"serves"| HTML5
  WZ -->|"OData V4, user token"| SRV
  SRV --> HDI
  SRV --> XS
  SRV --> DST
  SRV --> OBS
  DST -->|"tunnel"| CC --> GWQ
  DST -->|"OAuth 2.0 SAML bearer"| PC
  MAG -->|"OAuth 2.0 client credentials"| SRV
  SRV -.->|"HTTP via destination"| MAG
```

5.2 Customer Monitor application (purplecat)

Building block	Responsibility	Status
Fiori Elements app (List Report, Object Page, admin lists)	Work queue, request review and editing, parameter dialogs, "Run step" menu; served from the HTML5 Application Repository through the central launchpad	✅ (admin screen 💡)
`CustomerMonitorService` (CAP, OData V4)	Authorization (role + sales-org filter), drafts and locks, release check, all steps, backend routing, request log	✅ logic, 🟡 API calls
`IntakeService` (CAP, REST)	Three intake operations for Magento, idempotency on the Magento IDs, process-type derivation, approval check	🟡
Backend registry and SAP adapter	backendSystem → S/4 destination, Magento destination, adapter type, number rules; `API_BUSINESS_PARTNER` calls through the SAP Cloud SDK	🟡
Outbox worker	Sends ID notifications and rejections to the shops after the commit, retries with backoff, logs the result	🟡
HDI container (SAP HANA Cloud)	Requests, drafts, log, customizing, user assignments, Magento mapping, outbox	✅
Retention job (SAP Job Scheduling Service) and audit	Deletion of completed requests, audit of personal-data access	💡

logging (SAP Audit Log service)		

Screens:

Screen	Content
List (List Report)	Filter bar (search, region 💡, status group, released, in clarification, customer, sales org, process type, status, source, order exists, created/changed at/by); tabs My work / Errors / Clarification / Release / All with counters; columns status, hints (lock, clarification, order), release reason, process type, name, next step, waiting since, customer, sales org, Magento company, order volume, created, changed; errors highlighted red
Detail (Object Page)	Header: status, release, next step, key figures; sections Main data, Contact person *or* Ship-to (depending on process type), Sales data, Assignment, Log; actions: Edit, "Run step" menu (only allowed steps active), Refresh
Dialogs	Parameter dialogs for reject (reason), clarify (comment), assign / change assignment (business partner with value help); confirmation questions coming from the service
Admin 💡	Small Fiori Elements lists for customizing and user assignments per region (role `Kundenmonitor_Admin`)

The UI is SAP Fiori Elements (List Report and Object Page), so it follows the SAP Fiori design and accessibility standards. UI languages: English and German (more can be added). Times are shown in the user's local time zone.

Data displayed and acted on:

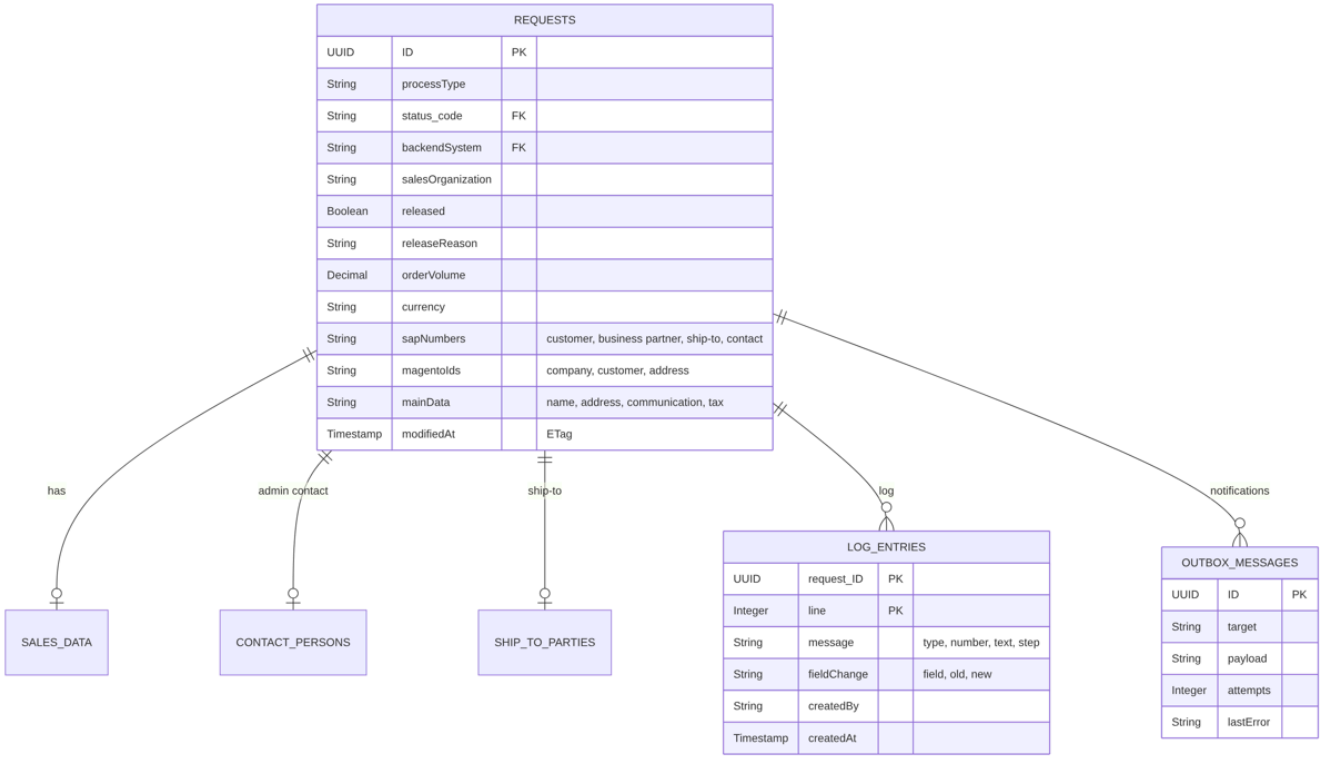


Diagram source (Mermaid)

03-data-model.mmd

```
erDiagram
  REQUESTS ||--o| SALES_DATA : "has"
  REQUESTS ||--o| CONTACT_PERSONS : "admin contact"
  REQUESTS ||--o| SHIP_TO_PARTIES : "ship-to"
  REQUESTS ||--o{ LOG_ENTRIES : "log"
  REQUESTS ||--o{ OUTBOX_MESSAGES : "notifications"
  REQUESTS {
    UUID ID PK
    String processType
    String status_code FK
    String backendSystem FK
    String salesOrganization
    Boolean released
    String releaseReason
    Decimal orderVolume
    String currency
    String sapNumbers "customer, business partner, ship-to, contact"
    String magentoIds "company, customer, address"
    String mainData "name, address, communication, tax"
    Timestamp modifiedAt "ETag"
  }
  LOG_ENTRIES {
    UUID request_ID PK
    Integer line PK
    String message "type, number, text, step"
    String fieldChange "field, old, new"
    String createdBy
    Timestamp createdAt
  }
  OUTBOX_MESSAGES {
    UUID ID PK
    String target
    String payload
    Integer attempts
    String lastError
```

```
36  }
```

Entity	Content	Source	Editable
Request (`Requests`)	Backend/region, process type, status, source, sales org; SAP numbers (customer, business partner, ship-to, contact person); Magento IDs (company, customer, address); order volume, currency, credit card; release (flag, procedure, reason, limit, by/at); clarification; main data (name 1–4, address, communication, VAT/tax numbers, industry, DUNS)	SAP HANA Cloud on BTP (today: `ZEC_CUSTOMER` + working copy `PAYLOAD_CHNG`); created by the intake, changed by the user and the steps	Main data only; numbers, status, release are read-only
Sales data (`SalesData`)	Sales org, distribution channel, division, sales group/office, customer group, sales district, Incoterms	Payload	Yes (except sales org)
Contact person (`ContactPersons`)	Shop role, title, name, function, department, language, phone, email	Payload	Yes (not for `CR_WE`)
Ship-to (`ShipToParties`)	Name, address, VAT number, "address used in order"	Payload	Yes (only `CR_WE`)

Log (`LogEntries`)	Message type/class/number/text, step, field, old/new value, user, time; messages returned by the SAP API; notification results	Monitor log in HANA Cloud	Read-only
Code lists	Statuses, status groups, process types, steps, release states and reasons (texts DE/EN, FR/IT where maintained)	Customizing tables in HANA Cloud per backend, initial load from `ZEC_CUST_*`, maintained in the admin screen (UC-13)	Read-only for users
Business partner value help	Number, name, customer, customer role, Magento company (from the request's backend only)	`API_BUSINESS_PARTNER` of the request's backend + Magento mapping in HANA Cloud	Read-only
Magento mapping 💡	Magento company / customer / address ↔ business partner / customer / contact / ship-to	HANA Cloud (today: field or table on the customer in S/4 ❓)	Written by the steps

Multi-currency (EUR, USD, AUD) is supported: the order volume and the release limit are always shown with their currency.

5.3 SAP BTP services (HYDAC platform)

What HYDAC's global account already provides. HYDAC's BTP usage export for October 2026 (`Subaccount_Details.xlsx` : 22 subaccounts, all in region eu10 on AWS or eu20 on Azure) shows the following subaccounts that matter for the monitor.

Subaccount (directory)	What it runs today	Relevance for the monitor

`DEV` (Development), CF space `dev`	Cloud Foundry Runtime (Free plan), Destination, HTML5 Application Repository, Identity Authentication, SAP Build Code, SAP Build Process Automation, AI Core, Document Information Extraction	**Recommended development subaccount** instead of FSM-DEV-CPI
`QAS`, CF space `space_qas`	Market Rates (Bring Your Own Rates)	Candidate for the QAS stage ❓
`CFLPDEV`, `CFLPPRD` (Workzone)	SAP Build Work Zone, standard edition (DEV and PRD launchpad), Destination	HYDAC's central launchpads: the monitor is integrated there as a content provider; no Work Zone subscription in the application subaccount
`GeneralServices` ("General Services f.e. HanaDB")	SAP HANA Cloud	Shared database instance; can be mapped to the monitor's space (same region)
`CTMS`	Cloud Transport Management, Destination	The cTMS tenant `ctms-hydac`
`FSM-DEV-CPI`, `FSM-PRD-CPI` (Integration)	Integration Suite (standard edition, messages), Destination	Integration only: API Management proxy and iFlows (section 8.9); not the place for the application
`IASIPS`	Identity Provisioning connectivity (Europe)	Identity Authentication and Identity Provisioning exist (section 8.2)
`HYDAC-AP-AUTOMATE`	SAP HANA Cloud (HYDAC USA accounts payable automation)	Shows that HYDAC USA application data already lives in region eu10
No application PRD subaccount, no subaccount outside Europe	–	PRD subaccount to be created; a regional AU deployment would be

		HYDAC's first non-EU subaccount (section 7.3)

The export lists consumed services. Free services such as XSUAA, Connectivity and Application Logging do not appear in it and are to be confirmed in the entitlements of `DEV`. The export also does not tell whether the shared HANA Cloud instance in `GeneralServices` has the capacity and the policy clearance to serve the monitor; that is a question for HYDAC's BTP admin.

Entitlements requested, mapped to the requirements of section 10.1:

Entitlement	Requirement	Why it is needed	Needed from	If not granted
Cloud Foundry Runtime (MEMORY), 2–4 GB	REQ-A, REQ-I	Runs the CAP service (≈ 256–512 MB per instance), the HDI deployer task and the content deployers during deployment. Headroom for zero-downtime redeploys.	First deployment	No deployment possible. `DEV` currently uses the **Free** plan; the monitor needs the Standard (MEMORY) plan quota of 2–4 GB in `DEV` ❓ (the Free plan's quota may not suffice).
SAP HANA Cloud (`hana-cloud` / `hana` + tools)	REQ-P	Database instance for the HDI container; **the system of record from phase 2**. Data volume stays in the low gigabytes, the smallest size suffices.	First deployment	No persistence at all. **Recommendation:** map the existing HANA Cloud instance in `GeneralServices` to the monitor's DEV (and QAS) space (instance mapping, same region eu10); then no `hana-`

				`cloud` entitlement is needed in `DEV`, only `hdi-shared`. HANA Cloud is the **main cost item** of the setup. For PRD: own instance or `GeneralServices` per HYDAC's policy ❓, with the backup retention set to HYDAC's policy.
SAP HANA Schemas & HDI Containers (`hana` / `hdi-shared`)	REQ-P	The application's own schema (container) in the HANA Cloud instance, deployed from the archive	First deployment	as above
Continuous Integration and Delivery (`cicd-app` / `default`)	REQ-D	Builds the archive on every push to Git, runs lint and service tests, deploys to DEV, uploads to cTMS	When Git is set up	Alternative: GitHub Actions + `cf` CLI (no entitlement, but a pipeline to maintain). Manual builds are possible but not reproducible. The export shows no CI/CD subscription; SAP Build Code

				(Standard) is active in `DEV` – ❓ check whether its bundle already covers the CI/CD service before a separate entitlement.
SAP Build Work Zone, standard edition (`SAPLaunchpad` / `standard`)	REQ-A	Entry point for business users: tile, role-based visibility, later navigation to the S/4 app "Manage Business Partners"	Before user tests	**Not needed in the application subaccount.** HYDAC runs central launchpads in `CFLPDEV` and `CFLPPRD` ; the monitor is added there as a content provider (a design-time and a runtime destination from the launchpad subaccount to the application subaccount; both subaccounts use the same identity provider; the app role collections stay in the application subaccount). ❓ Confirm that this is HYDAC's standard and

				which site serves QAS.
Cloud Transport Management (`transport` / `standard`)	REQ-D	Moves the same archive DEV → QAS → PRD with approval and audit trail	Before QAS exists	**Not an entitlement for the application subaccount:** the tenant `ctms-hydac` runs in subaccount `CTMS` . We need from there a transport node per target space, a service key of the `transport` instance for the CI/CD upload, and the TMS role collections (section 7.5). DEV alone works without it.

Sizing note. The 2–4 GB Cloud Foundry quota and the smallest HANA Cloud size are initial estimates from the data volume of the legacy tables and the number of deployment tasks, not measured production sizing. User counts, requests per day and the deployment topology (central or regional, section 7.3) are needed before HYDAC approves the final capacity and cost ❓ (section 1.4).

Services without a new entitlement (in use in `DEV` according to the export, or unmetered and therefore not in the export – ❓ to confirm in the entitlements of `DEV`):

Service	Requirement	Why	In the export
`xsuaa` (application)	REQ-A, REQ-S	Login, roles `Viewer` / `Editor` / `Releaser` / `Admin` , attribute `SalesOrg` ;	not listed (unmetered)

		technical client per Magento shop	
`destination` (lite)	REQ-A, REQ-I	Route UI → CAP; one destination per SAP system and per Magento shop (Q4H/GWQ, AU Public Cloud, shops EU/US/AU)	in use in `DEV`
`connectivity` (lite)	REQ-I	Cloud Connector tunnel from the CF app to the on-premise business-partner API (not needed for the Public Cloud)	not listed (unmetered)
`html5-apps-repo` (app-host, app-runtime)	REQ-A	Hosting the UI, consumed by the launchpad subaccount	app-host in use in `DEV`
`application-logs` / Cloud Logging	REQ-O	Central logs. CAP supports both. Our archive has an optional SAP Cloud Logging binding; without it logs stay available via `cf logs`.	not listed

SAP Cloud Identity Services exist: Identity Authentication is bound to `DEV` and Identity Provisioning connectivity runs in `IASIPS`. ❓ Confirm that this is the tenant for productive use and whether the AU Public Cloud shares it (section 8.2). Optional, recommended once personal data is on BTP (section 8.6): **SAP Audit Log service** (audit of personal-data access) and **SAP Job Scheduling Service** (retention job).

5.4 SAP systems (HYDAC SAP Basis, HYDAC Australia)

Aspect	On-premise S/4 (Q4H)	S/4HANA Cloud Public Edition (Australia)
Connection	Cloud Connector, principal propagation	Internet, no Cloud Connector; communication arrangement `SAP_COM_0008` (OAuth 2.0 SAML bearer for user propagation, or a communication user with certificate; method to validate in the AU tenant ❓)
API	`API_BUSINESS_PARTNER` (activated standard service)	`API_BUSINESS_PARTNER` – **the same API, the same adapter**. In the Public Cloud it is the only way to create customers from outside anyway: the released ABAP Cloud business-partner object cannot create customer roles (SAP note 3655149)
Custom code	None needed	None needed. **No ABAP Cloud development, no developer extensibility required**
Custom fields	Z fields on the customer today (Magento company, CRM number ❓)	Key-user custom fields, if HYDAC wants them visible in S/4; otherwise the mapping lives in the monitor (section 6.3)
Number ranges, grouping	❓ per sales org	❓ per AU configuration
Users	S/4 users mapped by email / user ID	Business users with a business role for business-partner maintenance; same IAS identity possible (section 8.2)

USA on RISE with SAP. The future US system is S/4HANA Cloud Private Edition, hosted by SAP under RISE. For the monitor it behaves like the on-premise column: `API_BUSINESS_PARTNER` is the same standard service, the connection goes through a Cloud Connector operated in the RISE landscape (set up by HYDAC Basis with SAP per the RISE contract), and principal propagation works the same way (certificate trust and `CERTRULE` in the RISE system). No custom code is needed there either. In the backend registry the USA is one more backend `US` with the destination `S4_US`; when the RISE system goes live, only the destination and the Cloud Connector resource change. ❓ Timeline of the RISE migration, and which system processes US registrations until then.

Australia is the same design. The AU Public Cloud is one more SAP destination (`SAP_COM_0008`, SAML bearer) and one more Magento destination in the backend registry (section 8.3). There is no ABAP Cloud development, no developer extensibility and no second backend implementation. The effort for Australia is configuration, authorizations and testing. What has to be agreed with Gareth in a short workshop ❓: scope and AU sales orgs, the AU Magento flow today, whether AU wants the review-before-create process at all (if Magento AU creates customers directly today, that is a business decision before any technical one), business-partner number ranges and grouping in the AU tenant, and the identity setup.

5.5 Magento (TechDivision)

What changes for Magento and when:

- **No change for Magento in phase 2** (development and tests on DEV/QAS with test shops or replayed payloads).
- **Change at cutover:** TechDivision switches each shop's endpoint and credentials from `ZEC_CUSTOMER_CREATE_SRV` to the intake service. The payload structure stays. Notifications then arrive from BTP instead of S/4 with the same content. What Magento reads back today via `ZEC_CUSTOMER_READ_SRV` after an ID notification is either included in the notification or read from the intake service (❓ TechDivision).
- Parallel run per shop: a shop is switched when its open requests have been migrated; the legacy services are retired after one release cycle.
- Coordination with TechDivision / Magento is needed for: address-level rejection, keeping `RejectCusomterSet` as is, the read-back, the switch plan, and the delivery guarantee: notifications are delivered **at least once** (a retry after a lost response can repeat a message), so Magento should treat a repeated notification for the same request as harmless ❓ (today's BAdI notification has no retry at all).
- ❓ Do the USA and Australia use separate Magento instances, and how do AU registrations reach SAP today?

6. Runtime View

6.1 Request Lifecycle: Process Types, Statuses and Steps

Process type	Meaning	Start	Typical end
`CR_DEB`	New company without SAP customer	000 New	CREATE → 999 Done
`CH_DEB`	Company with existing customer	000	AP_MODIFY → 999
`CR_AP`	New shop user for existing customer	000	AP_MODIFY → 999
`CH_AP`	Change of existing contact person	997 Automatically done	–
`CR_WE`	New delivery address	010	CREATE_WE → 999

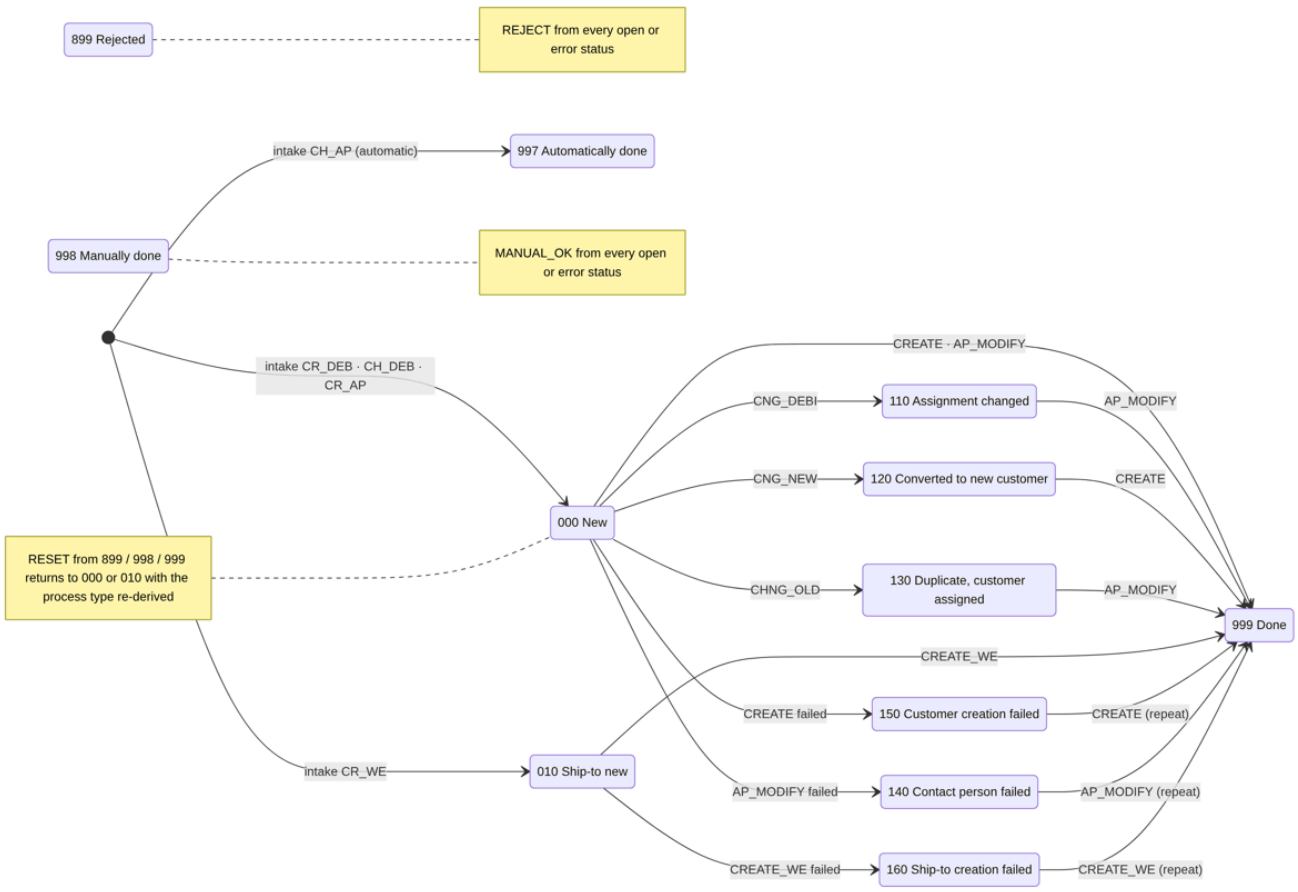


Diagram source (Mermaid)

02-status-transitions.mmd

```
stateDiagram-v2
  direction LR
  [*] --> s000 : intake CR_DEB · CH_DEB · CR_AP
  [*] --> s010 : intake CR_WE
  [*] --> s997 : intake CH_AP (automatic)
  s000 : 000 New
  s010 : 010 Ship-to new
  s110 : 110 Assignment changed
  s120 : 120 Converted to new customer
  s130 : 130 Duplicate, customer assigned
  s140 : 140 Contact person failed
  s150 : 150 Customer creation failed
  s160 : 160 Ship-to creation failed
  s899 : 899 Rejected
  s997 : 997 Automatically done
  s998 : 998 Manually done
  s999 : 999 Done
  s000 --> s130 : CHNG_OLD
  s000 --> s110 : CNG_DEBI
  s000 --> s120 : CNG_NEW
  s000 --> s999 : CREATE · AP_MODIFY
  s000 --> s150 : CREATE failed
  s000 --> s140 : AP_MODIFY failed
  s110 --> s999 : AP_MODIFY
  s120 --> s999 : CREATE
  s130 --> s999 : AP_MODIFY
  s010 --> s999 : CREATE_WE
  s010 --> s160 : CREATE_WE failed
  s150 --> s999 : CREATE (repeat)
  s140 --> s999 : AP_MODIFY (repeat)
  s160 --> s999 : CREATE_WE (repeat)
  note right of s899 : REJECT from every open or error status
  note right of s998 : MANUAL_OK from every open or error status
  note left of s000 : RESET from 899 / 998 / 999 returns to 000 or 010 with the process ty
```

Status group	Statuses	Meaning
Open	000, 010, 110, 120, 130	Waiting for the back office; the allowed steps depend on process type and status
Error	140, 150, 160	A master-data step failed; the SAP messages are in the log; the step can be repeated
Closed	899, 997, 998, 999	Rejected, done automatically, done manually, done; only RESET is offered

Steps: CREATE, CREATE_WE, AP_MODIFY, CHNG_OLD, CNG_DEBI, CNG_NEW, MANUAL_OK, REJECT, plus RELEASE, CLARIFY and RESET. Only CREATE, CREATE_WE and AP_MODIFY touch the SAP system; all other steps change the request in the monitor's database only. The allowed steps per process type and status, the release check (manual, credit card, order-volume

limit per sales organization) and the process-type changes (CHNG_OLD and CNG_DEBI → `CH_DEB`, CNG_NEW → `CR_DEB` with a fresh release check) are customizing data per backend, taken over from today's S/4 customizing tables. CLARIFY and RELEASE do not change the status.

6.2 Intake of a Registration (I-01)

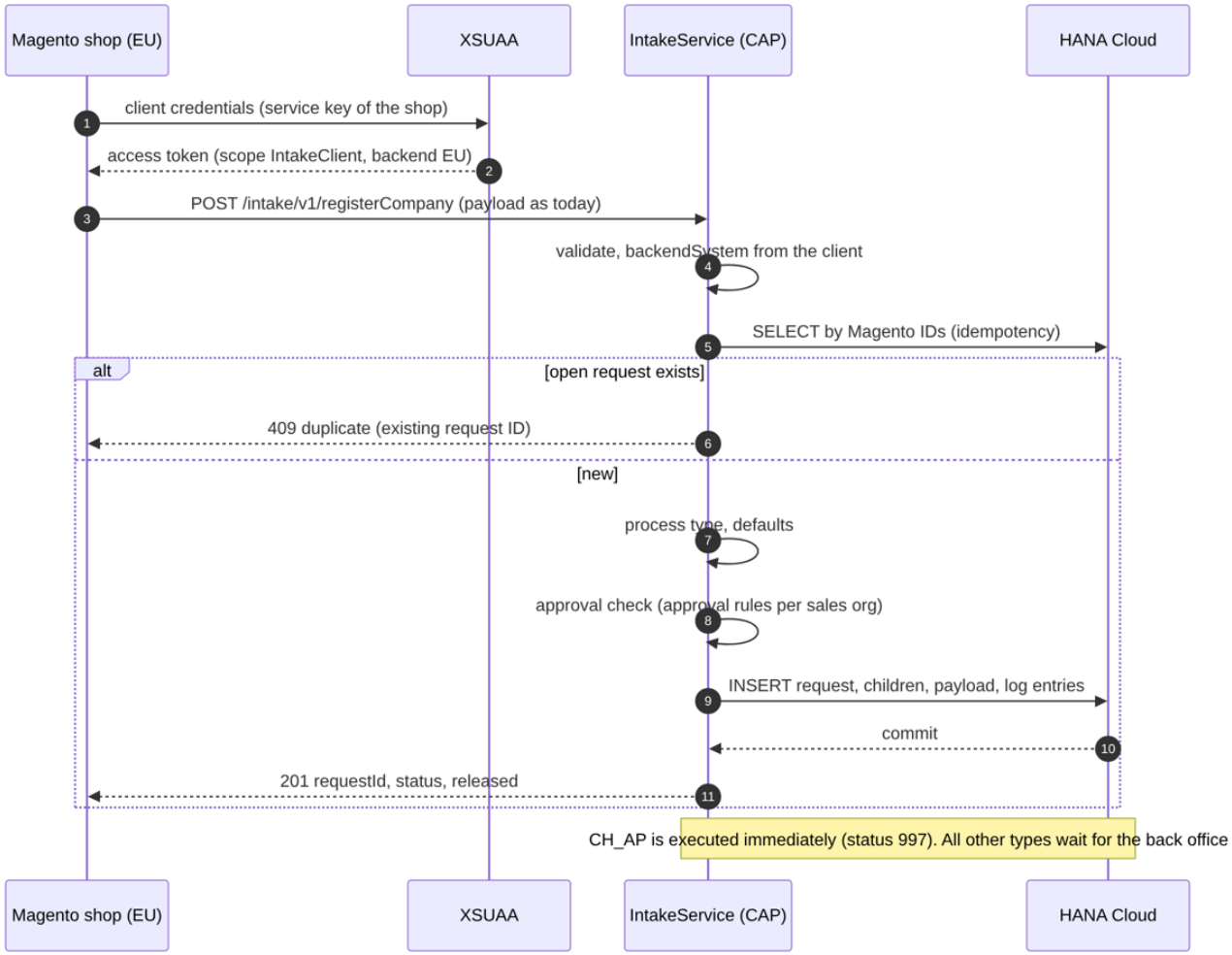


Diagram source (Mermaid)

11-intake-sequence.mmd

```
sequenceDiagram
  autonumber
  participant M as Magento shop (EU)
  participant X as XSUAA
  participant I as IntakeService (CAP)
  participant H as HANA Cloud
  M->>X: client credentials (service key of the shop)
  X-->>M: access token (scope IntakeClient, backend EU)
  M->>I: POST /intake/v1/registerCompany (payload as today)
  I->>I: validate, backendSystem from the client
  I->>H: SELECT by Magento IDs (idempotency)
  alt open request exists
    I-->>M: 409 duplicate (existing request ID)
  else new
    I->>I: process type, defaults
    I->>I: approval check (approval rules per sales org)
```

```
    I->>H: INSERT request, children, payload, log entries
    H-->>I: commit
    I-->>M: 201 requestId, status, released
  end
  Note over I,H: CH_AP is executed immediately (status 997). All other types wait for the
```

Error contract: 400 with a message list for missing mandatory fields, 401/403 for an invalid client, 409 for a duplicate, 503 if the database is unavailable (Magento retries). The payload structure is the one Magento sends to `ZEC_CUSTOMER_CREATE_SRV` today, so TechDivision changes only endpoint and credentials.

6.3 Master-Data Step CREATE against Q4H through GWQ (I-02)

How the steps map to the standard API:

Step	Calls on `API_BUSINESS_PARTNER`	Notes
CREATE (`CR_DEB`)	`POST A_BusinessPartner` (organization, `to_BusinessPartnerAddress`, `to_BusinessPartnerRole` FLCU00/FLCU01, `to_BusinessPartnerTax`) with deep insert `to_Customer` (`A_Customer` , `to_CustomerCompany` , `to_CustomerSalesArea`); then `POST A_BusinessPartnerContact` (+ `to_ContactAddress` , `to_ContactRelationship`) for the admin contact	Internal number assignment; ❓ grouping / number range per sales org. Sales org 7110 carrier as supplier role (`to_Supplier`) ❓ still needed
CREATE for an existing business partner without customer role	`POST .../to_BusinessPartnerRole` + `to_Customer`	

AP_MODIFY (`CR_AP` , `CH_DEB`)	`GET A_BusinessPartnerContact` ; `POST` or `PATCH A_BusinessPartnerContact` , `A_BPContactToAddress` , `A_BPContactToFuncAndDept`	Sales org 2000 "CRM number mandatory": ❓ which field; key-user custom field or rule retired
CREATE_WE (`CR_WE`)	`POST A_BusinessPartner` with `to_Customer` (ship-to as customer); then `POST A_CustSalesPartnerFunc` on the sold-to (partner function ship-to, sales area of the request)	New number persisted before the second call
CHNG_OLD, CNG_DEBI	`GET A_BusinessPartner('…')?$expand=to_Customer,to_BusinessPartnerRole`	`BP_NOT_FOUND` / `BP_NOT_CUSTOMER` ; Magento mapping read locally
Value help	`GET A_BusinessPartner?$filter=…&$expand=to_Customer`	Only in the request's backend, result limited
RELEASE, REJECT, CLARIFY, RESET, CNG_NEW, MANUAL_OK	none	Request in HANA Cloud only

What is validated and what is not 🟡. The call sequence above is designed from the API documentation of `API_BUSINESS_PARTNER` and the field usage of the legacy step classes; it has not yet been run against Q4H/GWQ or the AU tenant. The first connectivity tests on DEV have to confirm: the service activation and release level in Q4H, deep-insert support for business partner + customer + company code + sales area in one call, internal number assignment and grouping per sales org, the contact-person relationship, and the authorization checks on the propagated user. Each of these is a configuration or test question, not development.

A step end to end, here CREATE against Q4H through GWQ:

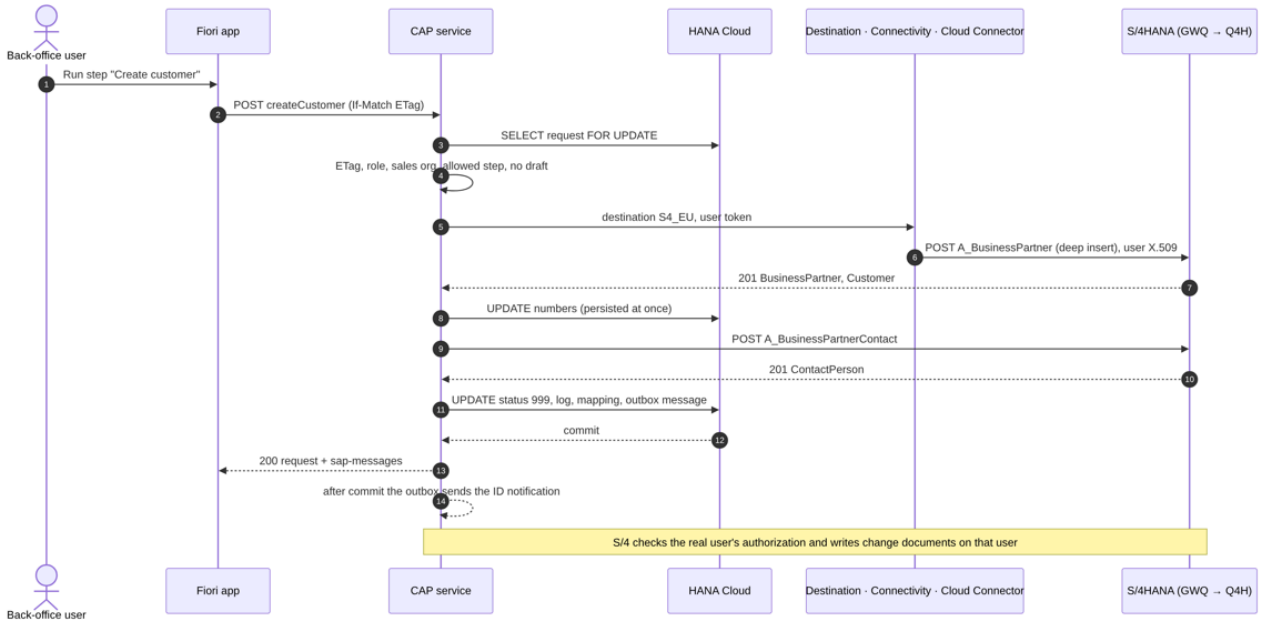


Diagram source (Mermaid)

09-create-step-sequence.mmd

```
sequenceDiagram
  autonumber
  actor U as Back-office user
  participant UI as Fiori app
  participant S as CAP service
  participant H as HANA Cloud
  participant C as Destination · Connectivity · Cloud Connector
  participant S4 as S/4HANA (GWQ → Q4H)
  U->>UI: Run step "Create customer"
  UI->>S: POST createCustomer (If-Match ETag)
  S->>H: SELECT request FOR UPDATE
  S->>S: ETag, role, sales org, allowed step, no draft
  S->>C: destination S4_EU, user token
  C->>S4: POST A_BusinessPartner (deep insert), user X.509
  S4-->>S: 201 BusinessPartner, Customer
  S->>H: UPDATE numbers (persisted at once)
  S->>S4: POST A_BusinessPartnerContact
  S4-->>S: 201 ContactPerson
  S->>H: UPDATE status 999, log, mapping, outbox message
  H-->>S: commit
  S-->>UI: 200 request + sap-messages
  S-->>S: after commit the outbox sends the ID notification
  Note over S,S4: S/4 checks the real user's authorization and writes change documents on
```

Failure paths: an API error sets the error status (140/150/160) and logs all messages, the created numbers stay on the request; a repeat continues with the missing part. An unreachable backend returns `BACKEND_UNAVAILABLE` and leaves the request unchanged. A stale version returns 412. A missing S/4 authorization of the propagated user returns 403 with the API message.

Magento mapping 💡. Today the Magento company hangs on the customer in S/4 (❓ Z field or table). The monitor keeps this mapping in HANA Cloud (written by the steps, migrated at

cutover). No Z field in S/4 is needed. If S/4 users must see the Magento company, a key-user custom field on the business partner can be written through the same API ❓.

6.4 Notification to Magento (I-03)

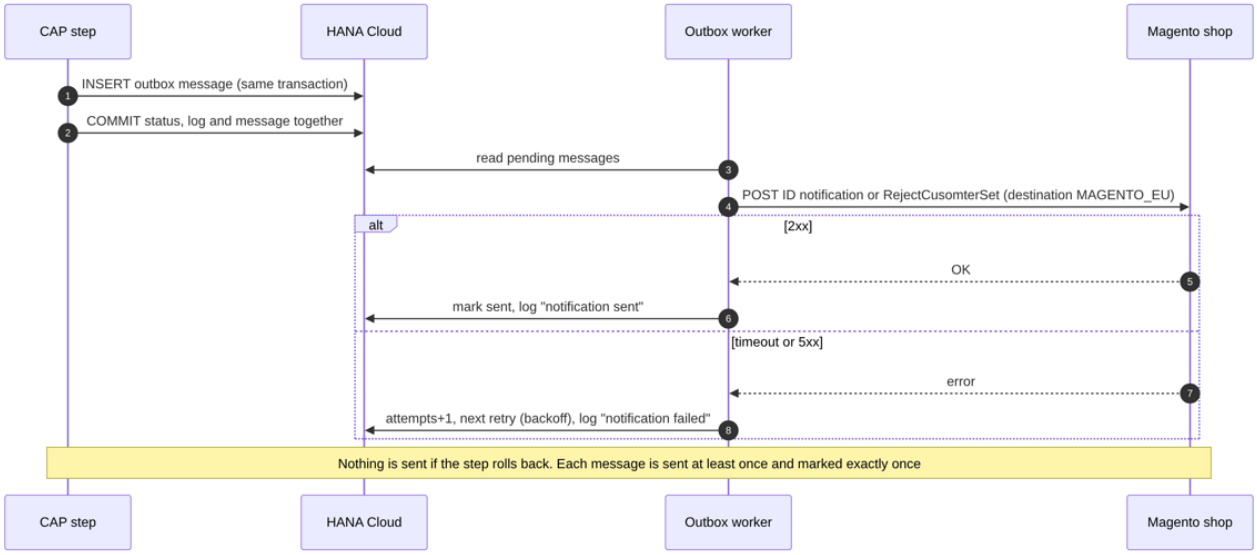


Diagram source (Mermaid)

12-notification-sequence.mmd

```
sequenceDiagram
  autonumber
  participant S as CAP step
  participant H as HANA Cloud
  participant Q as Outbox worker
  participant M as Magento shop
  S->>H: INSERT outbox message (same transaction)
  S->>H: COMMIT status, log and message together
  Q->>H: read pending messages
  Q->>M: POST ID notification or RejectCusomterSet (destination MAGENTO_EU)
  alt 2xx
    M-->>Q: OK
    Q->>H: mark sent, log "notification sent"
  else timeout or 5xx
    M-->>Q: error
    Q->>H: attempts+1, next retry (backoff), log "notification failed"
  end
  Note over S,M: Nothing is sent if the step rolls back. Each message is sent at least onc
```

A "resend notification" action for editors covers the case of a permanently failed message after the retries are exhausted.

7. Deployment View

7.1 Environments

Assumption: Q4H is the S/4HANA quality system (master data). GWQ is the SAP Gateway (hub) system of the same stage, where OData services are published (today

`ZEC_CUSTOMER_CREATE_SRV` to Magento; tomorrow `API_BUSINESS_PARTNER` to BTP). ❓ Please confirm, including the production pair (P-system / Gateway).

BTP stage	SAP EU (on-premise)	SAP USA	SAP AU (Public Cloud)	Magento
DEV (now)	Q4H via GWQ	❓ interim US system, or RISE test system once available	❓ AU test tenant	test shop / replayed payloads
QAS (later)	Q4H (❓ or a separate test client)	❓ RISE test system	❓ AU test tenant	staging shops
PRD (later)	❓ production S/4 + Gateway	RISE production system (❓ timeline; interim system until then)	❓ AU production tenant	production shops

The same archive is deployed on every stage: DEV in subaccount `DEV`, QAS in `QAS` ❓, PRD in a new subaccount (section 7.2). Only the destinations, which HYDAC's BTP admin maintains per subaccount, point to different systems and shops.

Region	SAP system (master data)	Connection	Status
EU	S/4HANA on-premise (Q4H/GWQ → production)	Cloud Connector	Interface design ready 🟡
USA	Own S/4HANA Cloud Private Edition on RISE with SAP (future)	Cloud Connector in the RISE landscape	Same design as EU (section 5.4); ❓ timeline, ❓ interim system
AU	S/4HANA Cloud Public Edition	Internet, `SAP_COM_0008`	Same design, no ABAP development (section 5.4)

Until the RISE system is available, the destination `S4_US` points at whichever system processes US registrations today ❓. If that is the EU S/4 system, US requests are simply more sales organizations reached through the EU connection until the migration; the switch to the RISE system is then a registry entry and a destination change, with no code change and no change for the users.

7.2 Subaccounts, Spaces and Platform Roles

Item	Requirement	Why	Note
Development subaccount: `DEV` (directory Development, eu10)	all	Place for the DEV space, service instances and subscriptions	`DEV` already runs Cloud Foundry, Destination, HTML5 Application Repository and Identity Authentication, so the monitor fits in with a new space `cap-monitor`. We withdraw the question about FSM-DEV-CPI: that subaccount is Integration Suite only. QAS could use the existing `QAS` subaccount ❓; a PRD subaccount for applications does not exist yet and has to be created. The region of the productive subaccount(s) depends on the data residency decision (section 7.3). HYDAC decides.

All subaccounts below are in HYDAC's global account, region eu10:

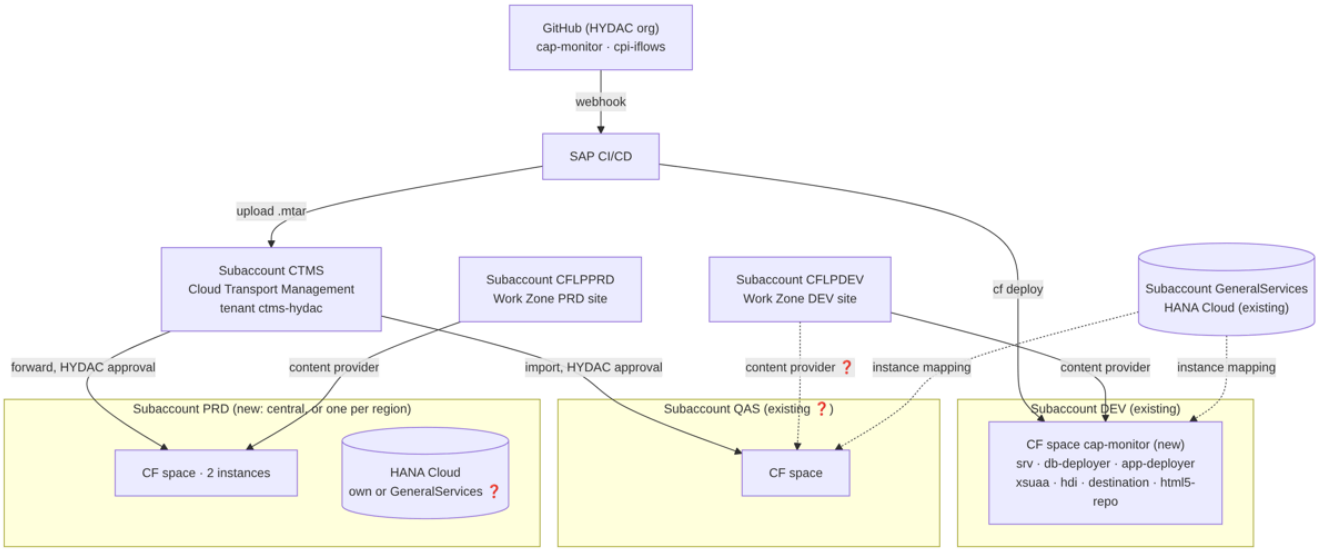


Diagram source (Mermaid)

05-subaccount-landscape.mmd

```
flowchart TB
  GH["GitHub (HYDAC org)<br/>cap-monitor · cpi-iflows"] -->|"webhook"| CICD["SAP CI/CD"]
  subgraph DEVSA["Subaccount DEV (existing)"]
    DEVSP["CF space cap-monitor (new)<br/>srv · db-deployer · app-deployer<br/>xsuaa · hdi
  end
  subgraph QASSA["Subaccount QAS (existing ❓)"]
    QASSP["CF space"]
  end
  subgraph PRDSA["Subaccount PRD (new: central, or one per region)"]
    PRDSP["CF space · 2 instances"]
    PRDHC[("HANA Cloud<br/>own or GeneralServices ❓")]
  end
  CTMS["Subaccount CTMS<br/>Cloud Transport Management<br/>tenant ctms-hydac"]
  WZD["Subaccount CFLPDEV<br/>Work Zone DEV site"]
  WZP["Subaccount CFLPPRD<br/>Work Zone PRD site"]
  GS[("Subaccount GeneralServices<br/>HANA Cloud (existing)")]
  CICD -->|"cf deploy"| DEVSP
  CICD -->|"upload .mtar"| CTMS
  CTMS -->|"import, HYDAC approval"| QASSP
  CTMS -->|"forward, HYDAC approval"| PRDSP
  WZD -->|"content provider"| DEVSP
  WZD -.->|"content provider ❓"| QASSP
  WZP -->|"content provider"| PRDSP
  GS -.->|"instance mapping"| DEVSP
  GS -.->|"instance mapping"| QASSP
```

Cloud Foundry space and platform roles:

Item	Requirement	Why	Least-privilege alternative
CF space `cap-monitor` (DEV)	all	Isolated runtime for the app and its service instances	–

Space Developer for purplecat developers	REQ-D	Deploy the archive, create service instances in the space, read logs	– (needed)
Org Manager	–	Only needed to create spaces or manage org members ourselves	**Not needed if HYDAC creates the spaces.** Space Manager is enough for adding team members to the space. We are happy to drop this.
Subaccount Service Administrator (role collection)	REQ-A, REQ-D	Create subscriptions (Work Zone, CI/CD, cTMS) and service instances in the cockpit	HYDAC's BTP admin does the one-time subscriptions and assigns the app role collections. After that Space Developer is enough for daily work.
App role collections `Kundenmonitor_*` for test users	REQ-S	Testing the role and sales-org behaviour	Assigned by HYDAC or by us; in production through IdP groups (section 8.2)
Work Zone `Launchpad_Admin` in `CFLPDEV`	REQ-A	Register the content provider, add the tile to the DEV site, assign roles	Can stay with HYDAC's launchpad team; we provide the destination values
CI/CD `CICD Service Administrator` / `Developer`	REQ-D	Configure the pipeline and credentials	–
cTMS roles (for example `TMS_Viewer`, `TMS_Export_Opera`	REQ-D	Upload archives and track transports; HYDAC controls imports into QAS/PRD	–

`tor` ; import roles stay with HYDAC)			

7.3 Hosting and Region (Open)

Central or regional deployment. The same archive supports one central deployment (one PRD subaccount, one HANA Cloud, all backends registered, one work queue) or one deployment per region (each region's requests stay in its BTP region, three times the HANA Cloud and operation cost). We recommend central, on the condition that HYDAC data protection confirms that Australian contact data may be stored in an EU BTP region (Australian Privacy Principle 8 restricts cross-border disclosure of personal information). Otherwise regional for AU and central for EU and USA. HYDAC's global account today has subaccounts only in Europe (eu10 on AWS, eu20 on Azure), and HYDAC USA's accounts-payable automation already keeps its data in eu10, so a central deployment for EU and USA follows existing practice; a regional AU deployment would be HYDAC's first subaccount outside Europe. ❓ The decision is needed before AU goes live, not before DEV.

With regional deployments, cTMS gets one PRD node per region and the same archive is imported into each (section 7.5). The hosting and region question of the shop integration itself is treated on the page "Australien-Konzeption: Integration Architecture" (section 7.3 there).

7.4 Connectivity, Cloud Connector and Destinations

Item	Requirement	Why
Confirmation that GWQ is exposed via Cloud Connector (or a destination for it)	REQ-I	The business-partner API will be called on the system where HYDAC publishes Gateway services (section 7.1)
Activation of the standard service `API_BUSINESS_PARTNER` in Q4H/GWQ (`/IWFND/MAINT_SERVICE`)	REQ-I	Standard SAP service, delivered with S/4HANA since 1709; no development. ❓ Release/SP of Q4H
Cloud Connector resource `/sap/opu/odata/sap/API_BUSINESS_PARTNER/` (path and all sub-paths)	REQ-I	The Cloud Connector only lets through the paths that are listed explicitly. **This is the only path the monitor needs.**

Authentication method	REQ-S	Principal propagation (recommended) or a technical user (section 8.1)
S/4 test users for purplecat developers in Q4H/GWQ with business-partner creation authorizations per sales org	REQ-I	Testing the steps with real authorizations. No development access (SE80/ADT) needed.

Cloud Connector. One resource per on-premise backend: virtual host `gwq-virtual:443` (HTTPS), path `/sap/opu/odata/sap/API_BUSINESS_PARTNER/` with all sub-paths. Nothing else. For the US RISE system the same single resource is configured on the Cloud Connector of the RISE landscape (connected to the same BTP subaccount, with its own location ID if HYDAC runs more than one Cloud Connector). Principal propagation additionally needs the Cloud Connector system certificate trusted in GWQ/Q4H (ICM), a certificate-to-user mapping rule (`CERTRULE` , for example on the email address) and the business-partner creation authorizations on the S/4 users of the editors. Viewers, releasers and admins need no S/4 user. No RFC resources are needed.

Destination	Type	Authentication	Used for
`S4_EU`	OnPremise (Cloud Connector), URL = virtual host of GWQ	`PrincipalPropagation` ; DEV start: `BasicAuthentication` with a technical user	I-02
`S4_US`	OnPremise (Cloud Connector of the RISE landscape, once the RISE system exists; interim system until then ❓)	`PrincipalPropagation` , like `S4_EU`	I-02
`S4_AU`	Internet	`OAuth2SAMLBearerAssertion` (audience, token service URL and client from the	I-02

		communication arrangement; name ID format email); alternative `ClientCertificateAuthentication` with a communication user	
`MAGENTO_EU`, `MAGENTO_US`, `MAGENTO_AU`	Internet	As configured in S/4 today (basic or token) ❓ TechDivision	I-03
`srv-api`	Internet (generated by the archive)	Token forwarding from the approuter	UI → CAP

7.5 Build, Delivery and Transport

Git:

Item	Requirement	Why
HYDAC-owned GitHub organization	REQ-V	HYDAC owns the code; Integration Suite's built-in Git integration only supports github.com
Repositories `cap-monitor` and `cpi-iflows`	REQ-V, REQ-D	One repository per lifecycle: the app is built by CI/CD, the iFlows (if any, section 8.9) are pushed and pulled by Integration Suite
Write access for purplecat developers	REQ-V	Daily development; branch protection and reviews as HYDAC requires
Personal access tokens (fine-grained, limited to these repositories, with expiry)	REQ-V	Integration Suite asks for a token on every push/pull

Read-only technical access + webhook to BTP CI/CD	REQ-D	Builds start automatically on push
Integration Suite tenant admin registers the repositories (Settings → Integrations → Git Access)	REQ-V	One-time setup per repository

Limitation: the built-in feature covers iFlows only, not message mappings or script collections. These stay in Integration Suite packages until SAP extends the feature.

Transport:

Item	Requirement	Why
Reuse of the cTMS tenant `ctms-hydac` in subaccount `CTMS` (eu10); confirm its owner	REQ-D	One transport tool for all HYDAC BTP content; the export confirms the tenant
In `CTMS`: transport node for the DEV space, later QAS and PRD nodes (each with a destination to the deploy service of the target space; with regional deployments one PRD node per region)	REQ-D	Same archive through all stages and regions; space-specific values (destinations, backends) via extension descriptors, never a rebuilt archive
Service key of the `transport` (standard) instance in `CTMS` for the CI/CD upload destination	REQ-D	CI/CD uploads the archive to the DEV node after each successful build
TMS role collections in `CTMS` for the team	REQ-D	Export/upload by purplecat, import approval by HYDAC

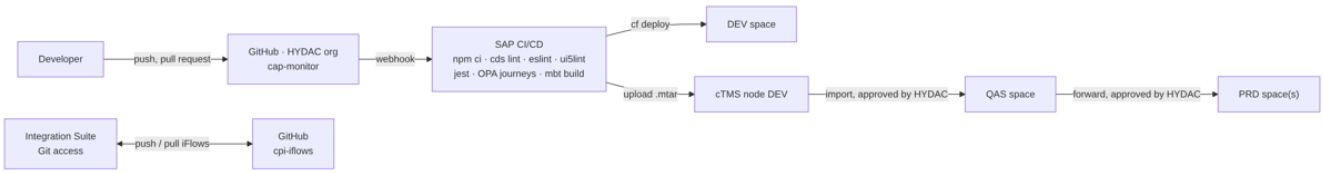


Diagram source (Mermaid)

06-build-and-transport.mmd

```
flowchart LR
  DEV["Developer"] -->|"push, pull request"| GH["GitHub · HYDAC org<br/>cap-monitor"]
  GH -->|"webhook"| CICD["SAP CI/CD<br/>npm ci · cds lint · eslint · ui5lint<br/>jest · OPA
  CICD -->|"cf deploy"| SDEV["DEV space"]
  CICD -->|"upload .mtar"| TMS["cTMS node DEV"]
  TMS -->|"import, approved by HYDAC"| SQAS["QAS space"]
  SQAS -->|"forward, approved by HYDAC"| SPRD["PRD space(s)"]
  IS["Integration Suite<br/>Git access"] <-->|"push / pull iFlows"| GH2["GitHub<br/>cpi-ifl
```

Aspect	Design
Build	`mbt build` produces one archive with the CAP service, HDI deployer, UI content and destination content
Quality gates	`cds lint`, ESLint, `ui5lint`; 137 service tests, 15 UI journeys in headless Chrome; all green before deploy ✅
Stages	The same archive moves DEV → QAS → PRD; stage-specific values (destinations, Cloud Logging binding, redirect URIs to the Work Zone host) come from extension descriptors, never from a rebuilt archive
Customizing	Data, not code: maintained in the admin screen per region, does not travel with the archive; the initial load comes from today's S/4 customizing

8. Cross-cutting Concepts

8.1 Authentication and Authorization

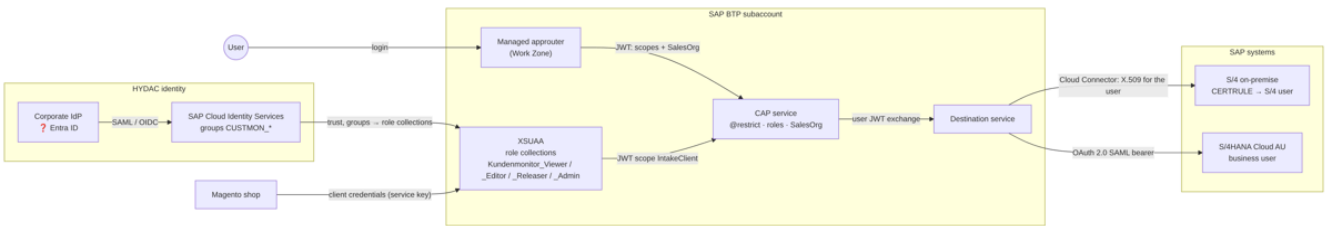


Diagram source (Mermaid)

10-authentication.mmd

```
flowchart LR
```

```
  subgraph HYDAC["HYDAC identity"]
    IDP["Corporate IdP<br/>❓ Entra ID"]
    IAS["SAP Cloud Identity Services<br/>groups CUSTMON_*"]
  end
  subgraph BTP["SAP BTP subaccount"]
    XS["XSUAA<br/>role collections<br/>Kundenmonitor_Viewer / _Editor / _Releaser / _Admin
    AR["Managed approuter<br/>(Work Zone)"]
    CAP["CAP service<br/>@restrict · roles · SalesOrg"]
    DST["Destination service"]
  end
  subgraph SAP["SAP systems"]
    S4["S/4 on-premise<br/>CERTRULE → S/4 user"]
    PC["S/4HANA Cloud AU<br/>business user"]
  end
  MAG["Magento shop"]
  IDP -->|"SAML / OIDC"| IAS -->|"trust, groups → role collections"| XS
  U((User)) -->|"login"| AR -->|"JWT: scopes + SalesOrg"| CAP
  CAP -->|"user JWT exchange"| DST
  DST -->|"Cloud Connector: X.509 for the user"| S4
  DST -->|"OAuth 2.0 SAML bearer"| PC
  MAG -->|"client credentials (service key)"| XS -->|"JWT scope IntakeClient"| CAP
```

Hop	Mechanism
User → Work Zone / app	HYDAC corporate identity provider via SAP Cloud Identity Services (❓ Entra ID?) → XSUAA; role collections from IdP groups (section 8.2)
UI → CAP	OAuth token forwarded by the managed approuter; CSRF protection by Fiori Elements
Magento → CAP (intake)	OAuth 2.0 client credentials: one XSUAA service key per shop/region (identifies the backend), rotated by HYDAC; optionally API key + quota in API Management
CAP → S/4 on-premise EU and S/4HANA Cloud Private Edition USA (RISE) (master-data steps, value help)	**Recommended: principal propagation.** The user's token is exchanged, the Cloud Connector issues a short-lived X.509 certificate for the logged-in user, and S/4 maps it to the S/4 user (rule in `CERTRULE`). Alternative: technical user (basic authentication)
CAP → S/4HANA Public Cloud	Communication arrangement `SAP_COM_0008` in the AU tenant; user propagation via OAuth 2.0 SAML bearer assertion; alternative client

	certificate with a communication user. ❓ To validate in the AU tenant: the supported authentication method of the arrangement, the business-user mapping and the business role for business-partner maintenance
CAP → Magento (notifications)	As S/4 does it today (basic authentication or token per shop; ❓ TechDivision), stored in the destination

Why principal propagation for the master-data steps: S/4 checks the authorization of the real user to create business partners for that sales org and company code; change documents and the created master data show the real user, not a technical one; there is no shared password to rotate. Effort for HYDAC: a one-time Cloud Connector trust setup (system certificate, CA), an ICM/certificate mapping in GWQ/Q4H, and a user mapping rule (for example email → S/4 user). Only editors who run CREATE / CREATE_WE / AP_MODIFY need an S/4 user.

Pragmatic start: a **technical user on DEV** for the first connectivity tests, then principal propagation **before user tests on QAS**. ❓ HYDAC decides. If HYDAC prefers a technical user permanently, S/4 cannot check per user who creates master data; the monitor's own authorization (role + sales org) is then the only check, and the real user appears in the request log only.

A user sees and changes only what all of these layers allow:

#	Layer	Where	Controls
1	Login	Corporate IdP → Identity Services → XSUAA	Only HYDAC users; the Work Zone tile is visible only with the Work Zone content role for the monitor
2	Function	Role collection (Viewer / Editor / Releaser / Admin)	Read, edit and run steps, release, maintain customizing
3	**Rows**	Sales-org assignment per backend (section 8.2), applied by the	Which requests are visible anywhere. This is the **single read**

		CAP service as a declarative restriction on every read: list, counters, detail, sub-data, log, value helps ✅	**path**: reads never touch S/4. A user with only AU sales orgs sees only Australian requests
4	Release	Releaser role	Unreleased requests are visible only to releasers ✅
5	**S/4 check**	Principal propagation on CREATE / CREATE_WE / AP_MODIFY	S/4 checks the creation authorization of the real user (sales org, company code) and writes the change documents on that user. No technical user can create master data anywhere

Why per backend: the same sales-org code (for example `1000`) can exist in two SAP systems with a different meaning, so the assignment carries the backend. ❓ If sales-org codes are unique across all systems, plain codes suffice.

8.2 User Management

Goal: no user-by-user maintenance in the BTP cockpit. Identity and roles come from HYDAC's corporate identity provider through SAP Cloud Identity Services (IAS): IAS groups (for example `CUSTMON_VIEWER`, `CUSTMON_EDITOR`, `CUSTMON_RELEASER`, `CUSTMON_ADMIN`) are mapped once to the role collections in the subaccount trust configuration. The **data scope** (sales organizations per user and backend) is maintained by regional key users in an admin screen of the monitor and stored in an application table; the service filter reads it on every request.

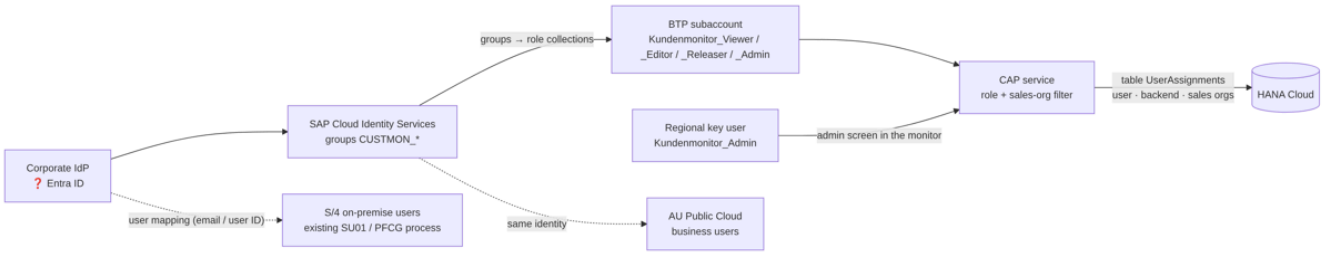


Diagram source (Mermaid)

07-user-management.mmd

```
flowchart LR
  IDP["Corporate IdP<br/>❓ Entra ID"] --> IAS["SAP Cloud Identity Services<br/>groups CUST
  IAS -->|"groups → role collections"| XS["BTP subaccount<br/>Kundenmonitor_Viewer / _Edito
  XS --> CAP["CAP service<br/>role + sales-org filter"]
  ADM["Regional key user<br/>Kundenmonitor_Admin"] -->|"admin screen in the monitor"| CAP
  CAP -->|"table UserAssignments<br/>user · backend · sales orgs"| HANA[("HANA Cloud")]
  IAS -.->|"same identity"| PC["AU Public Cloud<br/>business users"]
  IDP -.->|"user mapping (email / user ID)"| S4["S/4 on-premise users<br/>existing SU01 / P
```

Topic	How
Roles	IAS groups → role collections, standard BTP trust configuration; no API, no maintenance in the cockpit per user
Data scope	Application table `UserAssignments` (user, backend, sales organizations), maintained by regional key users in the admin screen; immediate effect; per-backend values and regional delegation are native; change log in the app, exportable for audits 💡
Alternative considered	Writing the scope as an IAS user attribute through the Identity Directory SCIM API from the monitor. Rejected as the default because the application would hold a tenant-wide `Manage Users` credential of an IAS tenant that is probably shared with other HYDAC applications; it takes effect only at the next login. Available if HYDAC wants every authorization visible in IAS ❓
Interim	Static role collections with sales orgs maintained in the BTP cockpit for the first test

	users on DEV
Joiner / mover / leaver	Identity and role groups in the IdP, HYDAC's normal process. A leaver loses access everywhere at once; group administration can be delegated per region
S/4 users	Principal propagation needs a matching S/4 user with business-partner authorizations **only for editors who run CREATE / CREATE_WE / AP_MODIFY**; viewers, releasers and admins need no S/4 user. **S/4 on-premise:** existing SU01 / PFCG process, mapped by email or user ID. **AU Public Cloud:** business users with a business role for business-partner maintenance restricted to the AU sales orgs; the Public Cloud uses IAS as well, so it can be the same identity (❓ same tenant as BTP?)
Consistency	One authorization matrix (role × region × sales org), owned by HYDAC, for IdP groups, the application scope and the S/4 roles. Out of sync means "no authorization" on the master-data step, never exposed data, because the service filter applies first. Quarterly review
Audit	BTP audit log (roles, trust), IdP/IAS logs (identity), application log (assignments and customizing changes), SAP Audit Log service (personal data 💡), S/4 change documents (master data, real user)

To confirm during the identity setup ❓: SAML or OIDC trust between the subaccount and IAS; which IAS tenant HYDAC uses and whether the AU Public Cloud shares it; the corporate identity provider (Entra ID?).

8.3 Multi-Backend Routing

Element	How it works

Backend per request	Every request carries the backend it belongs to (`backendSystem`: `EU`, `US`, `AU`), set by the intake from the shop's technical client; read-only; filter and column "Region" in the UI 💡
Backend registry	Configuration, not code: backend → S/4 destination, Magento destination, adapter type, number rules. A new region is one entry plus two destinations, no code change
Reads	Always from HANA Cloud: one list across all regions, filtered by the user's assignments. No remote reads, no synchronization
Writes	Every step and save changes the request in HANA Cloud; CREATE, CREATE_WE and AP_MODIFY additionally call the SAP system of the request as the real user
Rules per backend	Allowed steps, approval rules and texts are customizing tables per backend, maintained by the regional key users
Isolation	If one SAP system is unreachable, only that region's master-data steps fail with a clear message (`BACKEND_UNAVAILABLE`); everything else keeps working. If a shop is unreachable, its notifications wait in the outbox and are retried

8.4 Customizing per Region

Allowed steps per process type and status, approval rules and limits, and status and step texts are **data, not code**. They are stored per backend in HANA Cloud, loaded initially from today's S/4 customizing tables (`ZEC_CUST_*`) and maintained by the regional key users in the admin screen of the monitor (UC-13) 💡. A change takes effect immediately, without a deployment or a transport, and each region keeps its own rules.

8.5 Concurrency, Errors and Retry

Editing uses the Fiori draft with an edit lock, so two users cannot work on the same request at once. Every step carries the request's version (ETag) and is rejected with 412 if the request has changed in between; steps are serialized per request. The request, its log entries and the outbox message are written in one database transaction. The call to SAP is a separate operation: the numbers it returns (business partner, customer, ship-to, contact person) are persisted on the request at once, so a step that fails half-way can be repeated and continues with the missing part instead of creating a second business partner. An API error sets the error status 140/150/160 with all SAP messages in the log; an unreachable backend returns `BACKEND_UNAVAILABLE` and leaves the request unchanged; a missing S/4 authorization of the propagated user returns 403 with the API message. Notifications to Magento are sent only after the commit and retried with backoff (section 6.4).

8.6 Personal Data and Retention

Requests contain names, email addresses and phone numbers of contact persons. From phase 2 the requests are stored in HANA Cloud on BTP, which is then the **only** copy. The service marks these fields as personal data (CAP `@PersonalData`) and writes read and change audit entries to the SAP Audit Log service 💡. Completed requests are deleted by a scheduled retention job after a period to be defined by HYDAC ❓ (proposal: 24 months). Where the data is stored is a data residency question (section 7.3).

8.7 Operation, Monitoring and Backup

Signal	Source	Use
Health	`GET /health` (no login); the Cloud Foundry health check restarts the instance	Availability
Logs	Structured JSON with a correlation ID per request; optional SAP Cloud Logging binding for dashboards and alerts; authorization headers masked	Troubleshooting, audit trail of technical errors

Business KPIs	Counters of the list views (open, errors, clarification, release) inside the app; outbox status (pending, attempts, last error) 🟡; intake rejections 🟡	Daily operations by key users
Alerts	Cloud Logging alerts on error rate, outbox age, intake failures 🟡; BTP Alert Notification optional	Operations
Platform	BTP cockpit, `cf app`, HANA Cloud Central (memory, backups)	Capacity

Backup and continuity. HANA Cloud is managed by SAP: daily full backups and log backups at least every 15 minutes, point-in-time recovery, retention 14 days by default (configurable 1–215 days), to be set per HYDAC policy for PRD with a restore rehearsal once a year. The CAP service is stateless and redeployable from the archive at any time. Drafts are database rows and survive restarts. ❓ HYDAC's recovery objectives (how much data loss and how much downtime are acceptable, RPO/RTO) are not defined yet; they decide whether the managed backups suffice or a cross-region replica is needed.

8.8 Ongoing Administration and Ownership

The table lists the recurring tasks the application and its BTP services require. **"Owner" is for HYDAC to decide.** The proposal is only a starting point; its principle is simple: HYDAC owns identity, platform, SAP authorizations, production approvals and business rules; regional key users own everything at request level (sales-org assignments, customizing, first-level support); purplecat handles application changes and support as agreed; TechDivision owns the Magento side.

Area	Task	Frequency	Proposed owner	Owner (HYDAC decision)
Users & roles	Role group membership in the corporate IdP; sales-org assignment in the	On joiner / mover / leaver	Regional HYDAC key users (`Kundenmonitor_Admin`)	

	monitor's admin screen			
	IdP ↔ Identity Services ↔ BTP trust, group → role collection mapping, attribute mapping	One-time + on change	HYDAC IT (identity)	
	Authorization matrix role × region × sales org, periodic review	Quarterly	HYDAC IT + business owners	
Customizing	Process steps, approval rules and limits, status/step texts per region in the monitor's admin screen (no deployment, no transport)	On business change	Regional HYDAC key users (`Kundenmonitor_Admin`)	
Work Zone (`CFLPDEV` / `CFLPPRD`)	Content provider for the monitor, site, tile, roles; refresh the content provider after a deployment	Per release	HYDAC launchpad team	
Cloud Connector	Operation, upgrades, the one resource path, certificates (system + CA for	Quarterly / on expiry	HYDAC Basis	

	principal propagation)			
Destinations	One destination per SAP system and per Magento shop and stage; registry entry for a new region; credential rotation	On change / certificate or password rotation	HYDAC BTP admin	
SAP systems	`API_BUSINESS_PARTNER` available and authorized (business-partner creation per sales org), user mapping for principal propagation, number ranges; US (RISE): the same on the RISE system incl. its Cloud Connector; AU: communication arrangement `SAP_COM_0008`, communication system, business users and roles	On change / joiner-mover-leaver	HYDAC SAP Basis / security per region (US with SAP per the RISE contract); HYDAC Australia IT	

HANA Cloud (`GeneralServices`, PRD instance)	Instance size and mapping to the monitor's spaces, start/stop of non-productive instances, upgrades (SAP-managed, scheduled by the customer), backup retention (default 14 days) and a yearly restore test	Monthly check	HYDAC BTP admin	
Application	Monitor health, logs, alerts, outbox (stuck notifications), failed intakes; restart on incidents	Continuous (the health check restarts automatically)	purplecat (AMS) or HYDAC IT	
	Updates of CAP, UI5, Node.js runtime (security patches, SAP maintenance)	≈ quarterly	purplecat (development / AMS)	
	Deploy through cTMS: upload (DEV) / import approval (QAS, PRD per region)	Per release	purplecat upload, HYDAC approval	
CI/CD	Pipeline configuration, credentials, failed builds	On change	purplecat	

GitHub	Organization members, branch protection, token policy, offboarding	On change	HYDAC IT	
	Rotation of tokens (developer PATs, CI read token, Integration Suite)	On expiry (for example 90 days)	Token owner / HYDAC IT	
Magento	Intake credentials (service key per shop) and notification credentials, endpoint changes	On rotation / change	TechDivision with HYDAC eCommerce	
Integration Suite	API proxy for the intake (if used), legacy Magento iFlows, Git repository registration	Continuous / one-time	HYDAC integration team	
Business	First-level support for users, clarification of business rules	Continuous	HYDAC key users per region	
Data protection	Retention and deletion of completed requests (retention job),	Yearly / per policy	HYDAC data protection + IT	

	audit log review, data residency			
Cost & entitlements	Monitor consumption (HANA Cloud, CF memory, Work Zone), adjust quotas	Monthly	HYDAC IT	

What is **gone** compared with today: customizing transports in S/4, maintenance of the `ZEC_*` objects and Gateway services, the BAdI, and the monitoring of the Magento ↔ S/4 custom services.

8.9 Relationship to the Integration Suite iFlows

- **The monitor's runtime calls do not go through Integration Suite.** The master-data calls are synchronous user calls with user context (principal propagation); the notifications come from the outbox with their own retry. An iFlow in between would add latency and a second place to monitor without benefit.
- **Integration Suite is relevant for:**
  - an **API Management proxy** in front of the intake service, if HYDAC wants all external APIs (TechDivision is external) behind one gateway with API keys, quotas and analytics ❓;
  - a Cloud Integration iFlow only if a Magento shop cannot adapt its payload and a transformation is needed;
  - the Magento ↔ SAP flows that may be routed through Integration Suite today (❓ please confirm the current Magento ↔ GWQ and Magento ↔ AU paths);
  - later, replacing the legacy BAdI notification (customer maintenance with partner role ZO → Magento) with business-partner events from S/4 to BTP. That one is outside the monitor's scope today.
- **Version control:** iFlows go into the separate repository `cpi-iflows` in the same GitHub organization (section 7.5), so application and integration changes can be reviewed and traced together.

9. Architecture Decisions

ADR	Title	Essence	Status

ADR-01	Standard API only, no custom code in SAP	All master-data operations use `API_BUSINESS_PARTNER` of the respective SAP system; the `ZEC_*` objects are retired. Works identically for S/4 on-premise and S/4HANA Cloud Public Edition (section 5.4)	💡 Proposal
ADR-02	BTP is the system of record for requests	Requests, drafts, log, customizing, user assignments, Magento mapping and outbox live in SAP HANA Cloud; the SAP systems hold master data only (section 5.2)	💡 Proposal
ADR-03	One deployment, several backends	One UI, service and request store; per region one SAP and one Magento destination through a backend registry; central PRD deployment recommended, regional AU possible from the same archive (sections 7.3, 8.3)	💡 Proposal · ❓ data residency
ADR-04	Principal propagation for master-data steps	S/4 checks the real user and writes change documents on	💡 Proposal · ❓ HYDAC decides

		that user; technical user only for the first connectivity tests on DEV (section 8.1)	
ADR-05	Roles from the identity provider, data scope in the application	IAS groups → role collections; sales organizations per backend in the table `UserAssignments`, maintained by regional key users (section 8.2)	💡 Proposal
ADR-06	Notifications through a persistent outbox	Written in the step's transaction, sent after the commit, retried, at least once; "resend" action for editors (section 6.4)	🟡 Designed
ADR-07	Customizing as data per backend	Allowed steps, approval rules and texts in HANA Cloud, maintained in the admin screen, no transport (section 8.4)	💡 Proposal
ADR-08	One archive through all stages via cTMS	`mbt build` once, DEV → QAS → PRD through the existing tenant `ctms-hydac`; stage-specific values in extension descriptors (section 7.5)	💡 Proposal

10. Quality Requirements

10.1 Requirements

The requirements referenced throughout this document follow from the functional design in chapters 1 and 3. The last column is the index: where each requirement is answered.

Requirement	Meaning	Derived from	Answered in
REQ-A Access	Business users in EU, USA and AU open the app in the browser via SAP Build Work Zone, with roles per sales organization	UC-01, UC-02, section 1.4	5.3, 7.2, 8.1, 8.2
REQ-P Persistence	Requests, drafts, log, customizing, Magento mapping and outbox live in SAP HANA Cloud. The edit lock, the version check and the repeatable steps work on local rows	UC-03, UC-04, UC-12	5.2, 5.3, 8.5, 8.7
REQ-I Integration	Magento intake and notifications; master data through the business-partner API of each SAP system, Q4H/GWQ first	UC-05 to UC-08, UC-10, UC-14	3.3, 6, 7.1, 7.4
REQ-S Security	Authorization per role and sales organization; SAP authorizations and change documents on the real user for master-data creation	Sections 1.4, 8.1	7.2, 7.4, 8.1, 8.2
REQ-D Delivery	Reproducible builds, automated tests, controlled transport DEV → QAS → PRD	Chapter 7	5.3, 7.2, 7.5

REQ-O Operation	Health monitoring, logs, backups, handling of failed steps and notifications	Chapter 8	5.3, 6.4, 8.7, 8.8
REQ-V Version control	Source code and iFlows versioned and reviewable, owned by HYDAC	Chapter 7	7.5, 8.9

10.2 Tests and Validation

Stage	What is tested	Status
CI/CD on every push	`cds lint`, ESLint, `ui5lint`; 137 service tests (authorization, sales-org filter, release check, all steps, duplicate handling, error paths against mock backends); 15 UI journeys (OPA) in headless Chrome	✅
First connectivity tests on DEV (technical user)	Service activation and release level of `API_BUSINESS_PARTNER` in Q4H/GWQ; deep-insert support for business partner + customer + company code + sales area in one call; internal number assignment and grouping per sales org; contact-person relationship; Cloud Connector path	🟡 planned
Principal propagation on DEV/QAS	Certificate trust, `CERTRULE` mapping, authorization checks on the propagated user (403	🟡 planned, before user tests on QAS

	path), change documents on the real user	
AU tenant	Authentication method of `SAP_COM_0008`, business-user mapping, business role for business-partner maintenance, number ranges and grouping	🟡 planned, after the AU workshop ❓
US RISE system	Cloud Connector of the RISE landscape, `API_BUSINESS_PARTNER`, principal propagation and number ranges on the RISE test system – the EU test cases re-run against backend `US`	🟡 planned, once the RISE test system exists ❓
Magento (TechDivision)	Intake with replayed payloads and test shops; notification contract including repeated delivery; parallel run per shop before cutover	🟡 planned
User tests on QAS	Business users of each region with real S/4 authorizations; sizing figures ❓	🟡 planned

11. Risks and Open Questions (Cross-cutting)

All ❓ points of this document, collected for the review session. Risks are marked as such.

#	Topic	Question / risk	Addressee
Q01	Sizing	Number of users per role, region and sales organization; request volume per day per region (section 1.4)	HYDAC business owners, HYDAC IT

Q02	Landscape EU	Confirm the roles of Q4H (S/4 quality) and GWQ (Gateway hub) and the production pair (P-system / Gateway); QAS on Q4H or a separate test client (section 7.1)	HYDAC SAP Basis
Q03	Landscape USA	The USA will get an own S/4HANA system on RISE with SAP (S/4HANA Cloud Private Edition). ❓ Timeline of the migration, availability of a RISE test system for QAS, and which system processes US registrations until then (the EU S/4 with own sales orgs?) (sections 5.4, 7.1)	HYDAC Group IT, HYDAC USA
Q04	Australia	AU test and production tenants; scope and AU sales orgs; the AU Magento flow today; whether AU wants the review-before-create process at all; number ranges and grouping in the AU tenant; identity setup – short workshop with Gareth (section 5.4)	HYDAC Australia

Q05	Data residency	May Australian contact data be stored in an EU BTP region (Australian Privacy Principle 8)? Decides central vs. regional deployment for AU; needed before AU go-live (section 7.3). **Risk:** a late decision delays the AU go-live or doubles the operation cost	HYDAC data protection
Q06	Cloud Foundry quota	Standard (MEMORY) plan quota of 2–4 GB in `DEV`; the Free plan may not suffice (section 5.3)	HYDAC BTP admin
Q07	HANA Cloud	Can the shared instance in `GeneralServices` serve the monitor's DEV and QAS spaces (capacity, policy)? Own instance or `GeneralServices` for PRD? (section 5.3). **Risk:** a shared instance couples the monitor's performance and maintenance windows to other applications	HYDAC BTP admin
Q08	CI/CD	Does the SAP Build Code bundle in `DEV`	HYDAC BTP admin

		already cover the CI/CD service, or is a separate entitlement needed? Alternative GitHub Actions (section 5.3)	
Q09	Work Zone	Confirm that the central launchpads `CFLPDEV` / `CFLPPRD` with content providers are HYDAC's standard, and which site serves QAS (section 5.3)	HYDAC launchpad team
Q10	Subaccounts	Reuse of the `QAS` subaccount for the QAS stage; creation of the PRD subaccount(s) (section 7.2)	HYDAC BTP admin
Q11	Entitlements	Confirm the unmetered services (`xsuaa` , `connectivity` , `application-logs`) in the entitlements of `DEV` (section 5.3)	HYDAC BTP admin
Q12	cTMS	Confirm the owner of the tenant `ctms-hydac` in subaccount `CTMS` ; transport nodes and role collections (section 7.5)	HYDAC BTP admin

Q13	Identity	SAML or OIDC trust between the subaccount and IAS; which IAS tenant is productive and whether the AU Public Cloud shares it; the corporate identity provider (Entra ID?) (section 8.2)	HYDAC IT (identity)
Q14	Authentication to S/4	Principal propagation (recommended) or a technical user for the master-data steps (section 8.1). **Risk:** with a permanent technical user S/4 cannot check per user who creates master data, and change documents show the technical user	HYDAC IT (security), SAP Basis
Q15	Sales-org codes	Are sales-org codes unique across all SAP systems? If so, assignments without backend suffice (section 8.1)	HYDAC IT
Q16	Data scope in IAS	Should the sales-org scope additionally be visible in IAS (SCIM attribute), accepting a tenant-wide `Manage Users` credential in	HYDAC IT (identity)

		the application? (section 8.2)	
Q17	`API_BUSINESS_PARTNER` in Q4H	Release/SP of Q4H; activation of the service in `/IWFND/MAINT_SERVICE`; Cloud Connector exposure of GWQ (section 7.4). **Risk:** if the release does not support the deep insert of business partner + customer + sales area in one call, the step needs more calls, with the same persisted-numbers logic	HYDAC SAP Basis
Q18	Number ranges and grouping	Business-partner grouping / number range per sales org; is the supplier role (`to_Supplier`) for the sales org 7110 carrier still needed? (section 6.3)	HYDAC SD / business
Q19	Sales org 2000	"CRM number mandatory": which field, key-user custom field or rule retired? (section 6.3)	HYDAC business
Q20	Magento mapping in S/4	Where does the Magento company hang on the customer today (Z field or	HYDAC SAP Basis, business

		table)? Do S/4 users need it visible (key-user custom field)? (section 6.3)	
Q21	Notification contract	Credentials of the notification endpoints (basic or token); the read-back after the ID notification; address-level rejection for `CR_WE`; repeated notifications must be harmless (at least once) (sections 5.5, 7.4). **Risk:** Magento treats a repeated notification as a new event	TechDivision
Q22	Shop instances	Do the USA and Australia use separate Magento instances, and how do AU registrations reach SAP today? (section 5.5)	TechDivision, HYDAC eCommerce
Q23	Integration Suite today	Are the current Magento ↔ GWQ and Magento ↔ AU paths routed through Integration Suite? (section 8.9)	HYDAC integration team
Q24	API Management	Should the intake service sit behind an API Management proxy with API keys,	HYDAC integration team

		quotas and analytics? (section 8.9)	
Q25	Retention	Retention period for completed requests (proposal: 24 months) (section 8.6)	HYDAC data protection
Q26	Recovery objectives	RPO/RTO for the monitor; whether the managed HANA Cloud backups suffice or a cross-region replica is needed (section 8.7)	HYDAC IT
Q27	AU communication arrangement	Supported authentication method of `SAP_COM_0008` (SAML bearer or certificate), business-user mapping and business role (section 8.1)	HYDAC Australia
Q28	Audit and scheduling services	Adopt SAP Audit Log service and SAP Job Scheduling Service for personal-data audit and the retention job (section 5.3)	HYDAC IT
Q29	Legacy validation	The call sequence of section 6.3 has not been run against Q4H/GWQ or the AU tenant yet. **Risk:** assumptions from the API documentation (deep insert, contact relationship,	purplecat, HYDAC SAP Basis

		authorization on the propagated user) are confirmed only in the first connectivity tests (section 10.2)	

12. Glossary

Term	Meaning
Debitor	Customer master record in SAP (customer role of the business partner)
Ansprechpartner (AP)	Contact person of a customer; a shop user in Magento
Warenempfänger (WE)	Ship-to party; a delivery address of a customer, created as a customer with the partner function ship-to on the sold-to
Innendienst	Back office; the users who review and process requests per sales organization
Freigabe / Release	Approval of a request before it may be processed; rule per process type and sales org (manual, credit card, order-volume limit)
Klärung / Clarification	Status flag with a mandatory comment for a request that needs information from outside the monitor
Request	One registration, shop user or delivery address received from Magento, with the original payload and a working copy; the unit of work in the monitor
Process type	`CR_DEB`, `CH_DEB`, `CR_AP`, `CH_AP`, `CR_WE` – what the request is and which steps apply (section 6.1)
Step	An action on a request: CREATE, CREATE_WE, AP_MODIFY, CHNG_OLD, CNG_DEBI,

	CNG_NEW, MANUAL_OK, REJECT, RELEASE, CLARIFY, RESET. Only the first three touch the SAP system
Status group	Open (000–130), Error (140–160), Closed (899, 997, 998, 999)
Backend (`backendSystem`)	The region's SAP system a request belongs to: `EU` , `US` , `AU` ; set by the intake from the shop's technical client
Backend registry	Configuration table backend → S/4 destination, Magento destination, adapter type, number rules
Outbox	Table of notifications to Magento, written in the step's transaction and sent after the commit with retries
Draft / edit lock	Fiori Elements draft handling: a request in edit is locked for other users
ETag / version check	Every step carries the request's version; a stale version is rejected with HTTP 412
Principal propagation	Forwarding the logged-in user's identity to S/4: token exchange, short-lived X.509 certificate from the Cloud Connector, mapping to the S/4 user via `CERTRULE`
Cloud Connector	SAP component in HYDAC's data centre that tunnels calls from BTP to on-premise systems; exposes only the listed resource paths
Destination	BTP configuration object with URL and authentication for a target system (`S4_EU` , `S4_AU` , `MAGENTO_*`)
XSUAA	BTP authorization and trust service: role collections, scopes, attributes, service keys for technical clients

IAS	SAP Cloud Identity Services – Identity Authentication; the identity provider proxy between HYDAC's corporate IdP and BTP
Work Zone content provider	Registration of the application subaccount in HYDAC's central launchpad subaccount, so its apps and roles appear in the site
cTMS	SAP Cloud Transport Management; moves the archive DEV → QAS → PRD with approvals (tenant `ctms-hydac`)
HDI container	The application's own schema in SAP HANA Cloud, deployed from the archive
CAP	SAP Cloud Application Programming Model (Node.js) – the service framework of the monitor
Fiori Elements	SAPUI5 template-based UI (List Report, Object Page) driven by OData annotations
`API_BUSINESS_PARTNER`	SAP's standard OData V2 API for business partner, customer, supplier, contact person and their sales and company-code data
`SAP_COM_0008`	Communication scenario of S/4HANA Cloud Public Edition that exposes `API_BUSINESS_PARTNER`
Q4H / GWQ	HYDAC's S/4HANA quality system and its SAP Gateway hub (*assumption*, section 7.1)
RISE with SAP / S/4HANA Cloud Private Edition	SAP-hosted S/4HANA with the full on-premise code line; HYDAC USA's future system. For the monitor it is reached like the on-premise EU system: Cloud Connector in the RISE landscape, `API_BUSINESS_PARTNER`, principal propagation
Key user	Regional HYDAC user with the role `Kundenmonitor_Admin` who maintains

	customizing and sales-org assignments
Extension descriptor	Stage-specific values (destinations, bindings, URLs) applied to the same archive at deployment time
Archive (`.mtar`)	Multi-target application archive produced by `mbt build` , containing service, database, UI and destination content
Magento mapping	Relation Magento company / customer / address ↔ business partner / customer / contact / ship-to, kept in HANA Cloud
APP 8	Australian Privacy Principle 8 – cross-border disclosure of personal information
RPO / RTO	Recovery point objective (acceptable data loss) and recovery time objective (acceptable downtime)

Change Log

Version	Date	Author	Change
1.1	2026-11	Igor Vorobev	Takes HYDAC's BTP usage export of October 2026 (`Subaccount_Details.xlsx`) into account
1.2	2026-11-09	Igor Vorobev	Adds the summary, the delivery status, the validation caveats and the requirement index
Confluence v1	2026-11-09	Igor Vorobev	Page created from the document; diagrams rendered from

			Mermaid, sources attached
Confluence v2	2026-11-09	Igor Vorobev	Restructured along arc42 to match the page "Australien-Konzeption: Integration Architecture"; content unchanged, chapter numbers and cross-references updated; decision log, ADRs, open-question list and glossary added as indexes of the existing content
Confluence v3	2026-11-09	Igor Vorobev	USA: the future US system is an own S/4HANA on RISE with SAP (S/4HANA Cloud Private Edition) – backend `US` with the EU mechanism (Cloud Connector, `API_BUSINESS_PARTNER`, principal propagation); updated in sections 1.1, 2 (C11), 3.1, 4.2 (D-15), 5.4, 7.1, 7.4, 8.1, 8.8, 10.2, Q03 and the multi-backend diagram; timeline and interim system remain open